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Factors Signaling a Movie's Success: Forecasting the Profitability of a Film During the Early Production Stage Based on its Internal Characteristics

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Abstract

The dynamic and lucrative nature of the creative industries, particularly the film sector, requires a comprehensive understanding of the determinants of financial success. This study employs a quantitative approach using the methods of data mining, semantic modeling, thematic distribution analysis, machine learning techniques, and econometrics analysis to develop and evaluate a predictive model for estimating a film's box office performance based on internal production signals. By analyzing historical data from "The Numbers" database, the paper aims to identify key factors influencing a film's financial viability during early production stages. As a result, the main signals having a significant impact on a film's financial success at the box office, encompassing emotional content, thematic coherence, and production-related estimators such as genre, source material, star power, and release timing, were determined, culminating in the validation of predictive models like random forest and neural networks. The study contributes to empowering stakeholders in making informed decisions, advancing the field of film production analytics and resulting in a more data-driven and financially sustainable film production ecosystem. Limitations regarding data quality and model transparency, including possible future expansion of the model to incorporate external factors, additional metrics and data sources, are acknowledged, as well as future research agendas outlined to address these concerns and achieve a more comprehensive understanding of financial performance prediction in the film industry.

Keywords: Film industry, Financial performance, Predictive model, Machine learning, Internal production signals

JEL: C52, M21, L82

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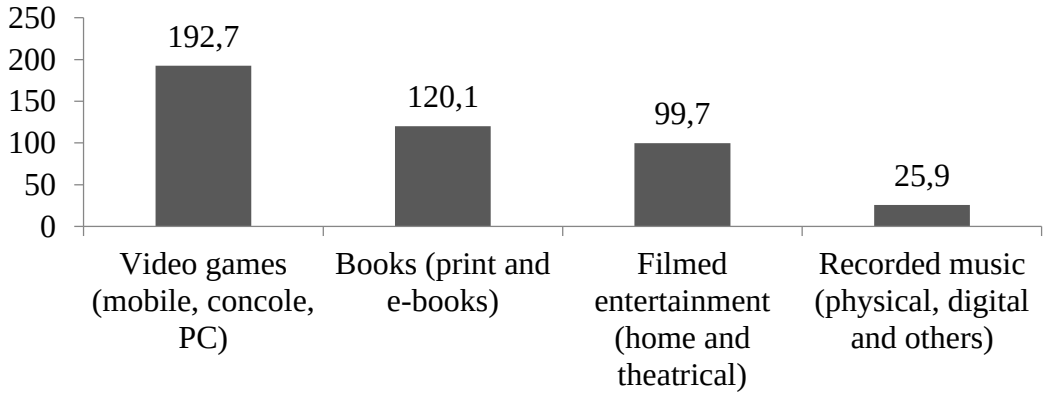
Introduction

The creative industries sphere is characterized by its appeal to individuals who think outside traditional boundaries, offering diverse forms of entertainment and art that allow people to unwind and transcend ordinary realities. This dynamic sector not only fulfills cultural and recreational needs but also serves as a highly profitable domain, attracting substantial investments due to its innovative nature and broad appeal. In accordance with the 2021 global statistics, movie entertainment industry was estimated to gain $99,7 billion, ranking third after video games and books (Richter, 2022) as presented in Figure 1. Understanding the factors that determine the financial performance of films, as a major driver of the global economy, is critical to making informed decisions and maximizing economic benefits. Additionally, given this context, the goal of the bachelor thesis is to analyze the prior findings and to develop a predictive tool can serve as a valuable asset for investors, aiding in risk mitigation and enhancing decision-making.

Figure 1. Estimated global revenue in billion $ from entertainment & media in 2021 (Richter, 2022)



This study considers a film as an asset with potential for investment. It is important for a film to attract investment before the project is fully ready. Academic literature suggests the time when the director and cast are announced to be the earliest possible phase for making a successful prediction of financial performance of a movie (Ahmed et al., 2020). Therefore, in order for a potential investor to be able to choose the most favorable investment options - pre-release stage movies, a need for a specific prediction tool occurs, that can be advantageous for both the investor and the film producers. Since it is very complicated to measure the impact of a film's external metrics, such as audience and critical acclaim, at the early stage of production - the period when investment and funds are most

required - we will focus on the impact of internal signals on the financial performance of films, which is the main delimitation of this study.

In addition, this study addresses the need for strategic analysis of film production. By identifying key internal factors contributing to financial success, filmmakers can refine their creative and business strategies, improving the likelihood of producing successful films. Therefore, the following research question is stated: **What factors in the early stage of film production determine the financial success of films at the box office?**

The issue of predicting the financial performance of films and the most affecting factors has been already covered in the global scientific literature, however, there exists a gap discussing the phenomenon of a “person power” - one of the movie success factors we use in the predictive model that we developed that includes the cumulative industry experience of the person involved in the film production process. Therefore, the present study attempts to extend the relevant empirical literature by addressing an early-stage film investment, which is an under-explored niche in the market.

The findings hold several essential managerial implications applicable to stakeholders across the film industry. Filmmakers can optimize strategic planning by leveraging data to target wider audiences, while distributors and producers can capitalize on successful films to sustain box office growth through culturally resonant marketing campaigns. Furthermore, government bodies and film funding agencies can use audience preferences to shape policies and foster innovation within the industry. Cinema owners can enhance the movie going experience by tailoring programming to popular films, fostering collaborations and promoting local productions. Investors gain valuable guidance for identifying opportunities and managing risks, reinforcing confidence in the long-term viability and growth potential of the cinema market.

The bachelor thesis is structured as follows. First, the paper analyzes the existing scientific literature, paying special attention to the theoretical foundations of signaling theory in the context of filmmaking, and also examines previous research methodologies, in particular qualitative approaches used in this field. Subsequently, the paper outlines the empirical methodology utilized for the current research and presents the acquired data. Lastly, the obtained results of the paper are stated and, consequently, the discussion and conclusion sections of the bachelor thesis cover insights into potential findings and their implications for the filmmaking industry, as well as the limitations of the current study.

Literature review

While considering investing in a movie, forecasting is crucial, as it enables stakeholders to anticipate potential financial returns, mitigate risks, and make informed decisions regarding resource allocation, production budgets, and marketing strategies. Hence, it is necessary to make a thorough review of relevant research papers related to the sphere as a basis for our paper. Based on these considerations, the Literature Review section is devoted to the analysis and comparison of previous studies. The literature review is organized as follows: first, the research examining the theory of the signaling theory in the context of the filmmaking industry was considered. Subsequently, the works focusing on various previously developed forecasting models, as well as factors most influencing movies financial results were reviewed. Finally, the papers based on the Internet data, Sentiment analysis and Word-of-mouth approaches are taken into consideration.

Signaling Theory within the Filmmaking Industry

Signaling theory, originating from information economics (Spence, 1973), serves as a crucial framework for understanding information asymmetry between buyers and sellers. Within the movie industry context, signaling theory elucidates how movie producers utilize quality signals to mitigate this asymmetry and influence consumer purchasing decisions (Fan et al., 2021). These signals play a pivotal role in shaping consumer perceptions, decision-making processes, and subsequent box-office performances. Additionally, the signals or factors further shaping the financial success of the movie can be considered by potential investors while deciding what projects are worth investing into. Signals can be broadly categorized into internal and external.

Internal signals encompass inherent product attributes such as star power, production budgets, genre, and storyline. These elements serve as indicators of quality and influence audience expectations and preferences. For instance, a high-profile cast or a substantial budget may signal to consumers that the movie is of high quality and worth their attention. Conversely, external signals originate from outside entities and include electronic word-of-mouth, reviews, ratings, and marketing campaigns results. These signals provide additional information to consumers, influencing their perceptions and decision-making processes. Positive reviews or word-of-mouth endorsements can enhance a movie's reputation and encourage audience engagement, while effective marketing campaigns can generate anticipation and excitement.

In our study, we focus primarily on the impact of internal signals on movies' financial performance. This focus is driven by the inherent challenges in measuring external indicators during the early stages of movie production when investments and funds are most critical. By examining how internal signals influence financial outcomes, we aim to provide valuable insights into the mechanisms through which movie producers can effectively signal quality and attract audiences, ultimately shaping the success of their films in the competitive marketplace.

Decision Support Systems in the Movie Industry. Forecasting Models Designed in the Prior Research

Decision Support Systems (DSS) play a critical role as the basis for investment related decisions in the film industry due to their inherent capacity to provide valuable insights in the initial stages of production. By leveraging DSS, stakeholders can make more informed decisions regarding which projects to invest in, mitigating financial risks and maximizing returns (Power, 2002), including the movie industry. Properly designed and using the relevant data, these systems offer a structured approach to evaluating factors such as cast popularity, director reputation, genre trends and other, ultimately enhancing the efficiency and effectiveness of investment strategies in the dynamic and competitive landscape of filmmaking. Eliashberg et al. (2009) present a practical case of DSS implementation through the SilverScreener system, demonstrating its effectiveness in optimizing revenue through movie scheduling. Despite occasional deviations from DSS recommendations, managerial decisions significantly contribute to net margin improvements, underscoring the system's robustness and flexibility. However, the study also emphasizes the importance of maintaining management confidence in the DSS and accommodating managerial needs in its development.

A number of authors claim that neural networks are popular tools for predicting the box-office success, showcasing the superior performance of these models compared to traditional statistical methods, achieving significantly better classification accuracy and application for the decision making (Sharda & Delen, 2006; Delen et al., 2007; Zhang et al., 2009). However, the reliance on neural networks raises questions about their interpretability and the potential biases inherent in data collection and model training. Additionally, the conversion of forecasting into a classification problem simplifies the complex dynamics of box-office success, potentially overlooking nuanced factors influencing audience preferences.

The article by the research group from Oklahoma State University, USA, continues the study in the field with the prediction of financial performance the movies of the Hollywood production based on the information fusion approach (Delen & Sharda, 2010). Results show that SVM outperformed other individual models, with ensembles generally performing better, particularly the fusion algorithm. The authors emphasize the potential applicability of their models beyond box-office predictions, highlighting the significance for decision-makers in the entertainment industry. In addition, the introduction of a pre-production forecasting model, featuring novel variables like production budget and pre-release advertising expenditures, highlights the algorithm's adaptability and effectiveness in capturing early-stage predictive factors (Ghiassi et al., 2015). Despite incorporation of promotional expenses, the model's accuracy remains consistent, indicating their limited impact on forecasting outcomes. Another work builds on a long-standing theoretical literature - Ahmed, Waqas, and Afzal (2020) address the challenge of pre-production box-office forecasting, emphasizing the importance of leveraging machine learning techniques and comprehensive datasets. While there is conclusive evidence on hybrid model demonstrating high accuracy rates, there is no consensus on the scalability and generalizability of their approach.

Overall, it is clear that decision support systems (DSS) serve as pivotal tools for enhancing investment decisions in the movie industry. Key factors such as production budget, advertising expenditures, star power, genre trends, technical effects, sequel status, and other ones significantly impact box-office revenues. Additionally, audience preferences, market dynamics, critical reception, and effective marketing strategies contribute to a movie's financial success. Consequently, it is argued that significance of DSS is critically important in enhancing decision-making processes and mitigating financial risks within the dynamic and competitive landscape of the film industry.

Sentiment Analysis and WOM Social Media Data Analysis for Enhancing Forecasting Results

In examining the various studies on forecasting movie financial performance, it becomes evident that sentiment analysis has become a focal point for researchers. A recent study conducted by Qiu and Zheng (2023) shed light on the prevailing issue of forecasters consistently underestimating or overestimating box office revenues, advocating for trimming strategies to address bias and variance. They explore two trimming approaches, one generating solely positive weights and another removing model associated with negative

weights before optimization. Despite resulting in slightly lower forecast accuracy compared to unconstrained weights, their findings hint at the potential for refining trimming strategies to improve overall forecasting accuracy. While trimming may mitigate bias and variance, it can also result in lower accuracy compared to unconstrained weights, potentially limiting its practical applicability.

An intriguing paper by a group of Russian researchers uses sentiment analysis techniques for films box office revenue forecasting, namely by examining the relationship between viewers' emotional experiences and box office receipts (Noakk et al., 2012). While factors analysis resulted in showing how emotions like sadness can notably impact revenue, their findings do not reveal a significant linear relationship between revenue and emotional evaluation, thus, the lack of a significant linear relationship challenges traditional forecasting methods. This prompts a reevaluation of how emotional factors are incorporated into forecasting models and the need for nuanced approaches.

The efficacy of integrating sentiment analysis with machine learning algorithms for box-office forecasting has been documented in the literature (Hur et al., 2016). Through experiments, the researchers demonstrate that audience-related factors, including sentiments expressed in reviews, play a crucial role in box-office forecasting both before and after a movie's release. Interestingly, neutral comments were found to be more influential before release, while emotional comments became more important post-release. However, the reliance on sentiment analysis raises questions about the robustness of such methods, particularly concerning the subjective nature of sentiment interpretation and potential biases in data collection and analysis.

Social media offers a rich source of data that can be used for sentiment analysis, allowing researchers and businesses to more accurately gauge public opinion, word-of-mouth (WOM) and predict outcomes based on trends and reactions in real time. The article explores the utilization of the data collected from the Twitter publications to forecast the performance of the movies shot in Bollywood, focusing on box office revenue and opening weekend collections (Gaikar & Marakarkandy, 2015), highlighting the significance of pre-release sentiment and hype, and demonstrates promising predictive accuracy in forecasting box office revenue and opening weekend collections. Continuing the research in box office prediction using social media content, Chinese researchers present an alternative approach to movie box office prediction by utilizing WOM data collected from microblogging and conducting sentiment analysis (Zhang, 2022). Their models demonstrate superior predictive power compared to traditional methods, emphasizing the importance of social media

analysis in forecasting movie performance. Nevertheless, the above studies' reliance on social media data may overlook other influential platforms or demographics, raising concerns about the representativeness and generalizability of findings.

Moving beyond sentiment analysis, Hofmann et al. (2017) delve into the roles of electronic Word-of-Mouth (eWOM), director and star power in shaping movie financial success, showing how commercially successful stars receive a significant box office bonus, with the fame of the star having a higher impact in the film future success comparing to artistically successful stars. Besides, by employing signaling theory, another study of the factors influencing box office performance in the Chinese movie industry investigates how these internal and external signals affect box office revenues over different stages of a movie's theatrical run, while also considering the moderating influence of the signaling environment (Fan et al., 2021). Additionally, the results of the study document that star power significantly affects box office receipts in the week of the premiere, emphasizing the importance of high-profile actors and directors in reaching audiences. However, the emphasis on star power and eWOM volume overlooks other potential factors contributing to movie success, such as storytelling, production quality, and marketing strategies.

Overall, while sentiment analysis and social media data offer valuable insights into audience behavior and movie performance, researchers must critically evaluate the limitations and potential biases inherent in these methods. Therefore, for reliable and valid forecasting in the film industry a holistic approach integrating multiple data sources and considering broader contextual factors is essential.

The paper aims at developing a tool to predict the factors most affecting a film's financial success for filmmakers and investors to be able to select films most worthy of investment.

Formulation of hypotheses

After analyzing the results of the prior research related to movie financing, we have formed a summary table representing the generalized findings of researchers with a focus on factors that play a key role in predicting the financial results of a motion picture. A table with summary results is presented in Appendix 1.

For further process of data collection and modelling we have considered all possible factors of financial success of the film through the prism of signaling theory. Based on the analysis performed, the factors or signals were divided into the semantic subcategories. The

groups of signals influencing the financial success of the film were identified according to the principle of systematization of various aspects of the creation and distribution of the film.

- The "Content Factors" group includes elements reflecting the main characteristics of the plot and content of the film, such as genre, topic, source, whether the film is a part of a franchise or not, and emotions, which can serve as key signals to attract the attention of viewers.
- "Production factors" cover the technical aspects of making a film, such as production method, film timing, acting, directorial power, and producer power that can influence the quality and success of a film at the stage of its creation.
- The “Commercial Factors” group focuses on aspects related to the marketing and distribution of the film, such as seasonality, budget, MPAA rating and country of release, which determine the potential profit and success of the film in the film industry market.

Consequently, following the scientific literature review and on the basis of factors groups identified earlier, the following research hypotheses were developed:

H1: Content Factors (Genre, Topic, Source, Franchise, Emotions) have a statistically significant effect on the financial performance of a movie at the box office.

H2: Production Factors (Production Method, Film Time, Acting Power, Directing Power, and Producer Power) have a statistically significant effect on the financial performance of a movie at the box office.

H3: Commercial Factors (Seasonality by month, Budget, MPAA rating, Release Countries (number)) have a statistically significant effect on the financial performance of a movie at the box office.

Methods and Data

The current study employs a quantitative approach with Machine Learning and Data Mining technics (François, 2008) to investigate the factors influencing a film's financial performance during its early production stages. This methodology leverages the power of data analysis to uncover patterns and relationships between film characteristics and their eventual box office success.

Rationale for Quantitative Approach

A quantitative approach offers several advantages in the context of our research.

Firstly, it allows us to analyze a large and comprehensive dataset from "The Numbers", a well-established database encompassing information on over 39,000 films. This extensive data pool provides a robust foundation for identifying potential correlations between film attributes and financial outcomes.

Secondly, quantitative techniques such as machine learning and data mining are adept at handling complex relationships between variables. This is crucial, as a film's financial success is likely influenced by a multitude of factors, some of which may be subtle or interconnected.

By employing these techniques, we can uncover hidden patterns within the data that might not be readily apparent through traditional analysis methods.

Research Design

The following steps, which formed the basis of our entire research, were employed:

1. **Data Collection and Preprocessing:**
   - ***Source:*** The research will utilize data from "The Numbers", a comprehensive online database encompassing information on over 50,000 films. By leveraging this extensive database, we aim to gain a multifaceted understanding of the factors potentially impacting a film's financial trajectory.
   - ***Method:*** Web scraping techniques were employed to extract relevant data points from the website. A custom Python script was developed to navigate the website's structure and efficiently gather information while adhering to the website's terms of service.
   - ***Data Volume:*** 39,072 films were included in the collected raw dataset.
2. **Preprocessing Stages:**

- ***Initial Cleaning:*** Python, a versatile programming language, was employed for the initial stages of data parsing and cleaning. This involved extracting relevant information, addressing inconsistencies or missing values, and potentially applying techniques like data type conversion, outlier handling, and null value imputation.
- ***Advanced Manipulation:*** Following the initial cleaning with Python, R, a powerful software environment for statistical computing and graphics, takes center stage. R provides a robust suite of tools for further data manipulation and feature engineering. This stage involves transforming and refining the data to create features suitable for applying machine learning algorithms, such as text processing, feature scaling, and dimensionality reduction.

3. **Machine Learning Model Development:**
   - ***Preprocessed Data Utilization:*** The preprocessed data served as the foundation for training and evaluating various machine learning models. These models aimed to identify the key factors influencing a film's financial performance.
   - ***Specific Techniques:***

     o *Sentiment Analysis:* Techniques were employed to analyze the emotional tone of film synopses. This analysis can potentially reveal insights into audience reception and potential box office appeal.

     o Topics detection and Clustering: Nets and Hierarchical Clustering were used to represent the film groups based on their keywords, genres and other textual data points. These film embeddings could then be leveraged to arrange films based on thematic similarities, potentially revealing underlying patterns in audience preferences.

     o Machine Learning Algorithm Exploration: A range of machine learning algorithms were explored to identify the most effective model for predicting financial success. This exploration may have encompassed algorithms like regression models (e.g., linear regression, random forest regression), decision trees, and neural networks. By comparing the performance of these models through metrics like accuracy, precision, and recall, or calculated error metrics the most accurate and reliable model for prediction could be selected.

The application of these diverse techniques will allow us to construct a robust predictive model capable of assessing a film's early-stage potential for financial success. The following sections will delve deeper into the specific data collected and the algorithms employed for analysis.

Data Collection and Description

Data Source Description

The Numbers, established in 1997, offers a reliable and extensive source of film industry data, including information on over 50,000 films and 200,000 individuals associated with film production (Nash Information Services, n.d.)[1]. The database's unique approach of acquiring information directly from film studios ensures the accuracy and timeliness of data, particularly concerning box office performance. This makes The Numbers an ideal source for research projects, providing researchers with access to comprehensive and reliable data on film production personnel, box office performance, and film descriptions.

Film's data includes various categories, such as:

- **Box Office**: detailed information about a film's box office performance in the domestic country and worldwide, including a breakdown by weekend, day, and territory;
- **Cast and Crew**: a comprehensive list of cast and crew members, categorized by their roles (leading cast, ensemble, supporting cast, narrator, and production & technical credits);
- **Trailer**: a link to the film's trailer;
- **Synopsis**: a brief and detailed description of the film's plot;
- **Keywords**: a list of relevant keywords associated with the film, such as genre, themes, or stylistic elements;
- **Production Information**: information about the production company, country of origin, language, release date, etc.;
- **Ratings and Milestones**: a comparison of the film to other films based on various metrics, such as box office performance or critic ratings.

[1] Nash Information Services. (n.d.). About Us. The Numbers: https://www.the-numbers.com/about-us.

Data Collection: Web Scraping and Primary Data Acquisition

We decided to use web scraping as the primary method of data collection. Web scraping is the automated process of extracting information from websites using algorithms or specialized software programs known as web crawlers or bots (Lotfi et al., 2021). These programs allow researchers to navigate websites and gather specific data. The origins of web scraping can be traced back to the early days of the internet, in the 1990s (Gray, 1993). At first, web scraping was mainly used for commercial purposes such as price comparisons and market research. However, soon after, its potential for scientific research became apparent, and it has since become a vital tool for data collection within various fields.

Web scraped data can be considered as primary data if it meets the following criteria (Ten Bosch et al., 2018):

- **Directness:** The data is collected from the original source, in this case websites, without the need for any intermediaries.
- **Accuracy:** The data has been collected in a reliable and accurate way, ensuring its suitability for research purposes.
- **Relevance:** The data relates to the research question and supports the study objectives.

There are two primary types of web scraping: HTML-based and API-based. HTML-based extraction extracts data by analyzing the HTML code on a website, while API-based uses the website's Application Programming Interface (API) to request information in a machine-readable form. For our research, we primarily use HTML-based web scraping because of the specific structure of the websites we target. This approach allows us to accurately and efficiently extract the data we need, using our own algorithm to collect information from the site, written in Python.

The collected raw data contains information about 39,000 films and includes the following description:

- **Rank:** The film's position relative to other films in terms of box office revenue.
- **Year:** The year of the film's release.
- **Title:** The official name of the film.
- **Worldwide Box Office:** The total box office revenue generated by the film on a global scale.
- **Domestic Box Office:** The box office revenue generated by the film within the United States.

- **International Box Office:** The box office revenue generated by the film outside of the United States.
- **Synopsis:** A brief summary of the film's plot.
- **Opening Weekend:** The box office revenue generated by the film during its opening weekend.
- **Theatrical Run:** The number of weeks the film was screened in theaters.
- **Domestic Share:** The percentage of the film's total box office revenue that was generated within the domestic country.
- **Production Budget:** The estimated cost of producing the film.
- **Theater Counts:** The number of theaters in which the film was screened.
- **Domestic Adjusted Box Office:** The box office revenue generated by the film within the United States, adjusted for inflation.
- **Domestic Releases:** The number of times the film was released in theaters within the United States.
- **MPAA Rating:** The film's rating from the Motion Picture Association of America.
- **Running Time:** The length of the film in minutes.
- **Franchise:** Whether the film is part of a larger franchise.
- **Keywords:** A list of words and phrases that describe the film.
- **Source:** The link to the source of the data.
- **Genre:** The film's genre.
- **Production Method:** The method used to produce the film.
- **Production Companies:** The companies that produced the film.
- **Production Countries:** The countries in which the film was produced.
- **Languages:** The languages spoken in the film.
- **Cast & Crew:** A list of the actors and crew members who worked on the film.

The data collected will be utilized in our research to identify the factors that influence a film's financial performance and to construct a predictive model. The entire process of data preprocessing and manipulation will be described in the following section.

Preprocessing the data

During the data preprocessing process, several key steps were taken to ensure the accuracy and structured nature of the data.

Initially, data regarding the cast and crew were transformed from string representations into structured data formats, facilitating further analysis. This stage involved parsing the data using the `ast` library, splitting the information into individual components, and creating new columns based on these components.

Subsequently, data cleansing was performed. Extraneous characters were removed from columns containing financial data, such as "Worldwide Box Office" and "Domestic Box Office", and numerical values were extracted from string data, employing regular expressions and the Pandas library.

Data Calculus

Powers estimation

The process of calculating the power of the cast and crew involved several principles aimed at accurately quantifying their contributions. Initially, the data pertaining to the composition of the cast and crew underwent transformation from string representations into structured data formats. This transformation facilitated subsequent analysis by parsing the data using the ast library, segmenting the information into discrete components, and creating new columns based on these components.

Subsequently, the calculation of power was guided by several principles. The power of the leading cast members was determined based on the revenue group of the film and their individual importance, with a distinction made between those deemed 'IMPORTANT' and others. Similarly, the power of lead ensemble members, supporting cast, cameo appearances, unnamed roles, uncategorized roles, narrators, extras, documentary subjects, and interviewees was calculated using analogous principles. In each case, the revenue group of the film was considered alongside the significance of the individual's contribution, with a higher weight assigned to those labeled as 'IMPORTANT'.

The process involved splitting the cast and crew entries into separate components, extracting the names of individuals, assigning power scores based on predefined criteria, and aggregating these scores to obtain a cumulative measure of their influence. This cumulative measure was then integrated into the main dataset, enabling subsequent analysis and interpretation of the data.

Similar principles guided our approach to calculating the power of film producers and the countries as the film producers.

Countries' effect

We determined the distribution reach of each movie, quantified by the number of countries where the movie was released. The computation involves counting the occurrences of commas in the "Countries" column, representing multiple countries, and adding 1 to include the initial country. The resulting count, stored in the variable "N_countries" provides insights into the global distribution reach of the movies.

Also we calculated the power of each country as film producers, following the same calculus logic, as Cast & Crew have.

Franchises detection

An examination of film data was undertaken to discern affiliations with film franchises. This process entails extracting pertinent details, such as rating, film title, release date, and franchise association. Subsequently, the data is arranged based on release dates in descending order. By appraising the presence of franchise associations, a binary indicator termed "is_franchise" is generated, facilitating the differentiation between franchise and non-franchise films. Additionally, the code computes the aggregate count of films within each franchise, denoted by the variable "n_franchise".

Sentiment analysis

To enhance the understanding of emotional nuances within textual data, sentiment analysis techniques are employed, including the utilization of lexicons or dictionaries that assign numerical sentiment scores to words or phrases based on prior linguistic investigations. The NRC Word-Emotion Association Lexicon, developed by Saif M. Mohammad from the National Research Council Canada (NRC), serves as a significant resource in sentiment analysis endeavors. This lexicon encompasses both positive and negative sentiment values alongside eight emotional categories, derived through the manual annotation of a dataset using the Maximum Difference Scaling technique (Saif, 2010). With 14,182 unigrams classified as positive or negative and delineations of words' connections to emotions such as anger, anticipation, disgust, fear, joy, sadness, surprise, and trust, the NRC lexicon facilitates comprehensive sentiment analysis across diverse textual datasets. Its availability in multiple languages, supported by automatic translation, further broadens its applicability. Furthermore, the licensing terms of the NRC lexicon permit its unrestricted

usage for research purposes, providing researchers with valuable resources for investigating emotional dimensions within textual data.

In this study, sentiment analysis was conducted on textual data using the syuzhet package in R (Isasi, 2021). Specifically, sentiment scores and emotional associations were computed for a set of texts. The textual data consisted of descriptions of various films, each encapsulating narrative elements and thematic content. The syuzhet package facilitated the sentiment analysis process by employing pre-compiled lexicons or dictionaries, such as the NRC Word-Emotion Association Lexicon, which assigns numerical sentiment scores to words or phrases based on previous linguistic research. By leveraging the get_sentiment() and get_nrc_values() functions from the syuzhet package, sentiment scores and emotional associations were extracted from the textual data. These scores and associations provide insights into the emotional dimensions and sentiment polarity expressed within the film descriptions, thus contributing to a deeper understanding of the underlying emotional content conveyed in the texts.

Film classification

The classification of movies was conducted through a series of steps aimed at identifying thematic clusters based on keyword associations extracted from movie synopses. Initially, a subset of data containing movie titles, release years, synopses, MPAA ratings, and keywords was selected by the presence of word "alien" to suppress the data volume for faster calculations. The keyword column was parsed to extract individual keywords, and then each movie was represented as a binary vector indicating the presence or absence of each keyword. Subsequently, a network analysis and hierarchical clustering were performed to identify clusters of movies based on keyword associations.

The network was constructed using a bipartite graph, where movies and keywords were represented as nodes, and edges indicated the presence of a keyword in a movie text batch. Several community detection algorithms were applied to the network to identify cohesive clusters of movies. These algorithms included edge betweenness, fast-greedy, and walktrap methods, each offering unique approaches to identifying community structures within the network (Zhang et al., 2018). The quality of the clustering was assessed using modularity, a measure of the extent to which the network is divided into clearly delineated communities.

Furthermore, hierarchical clustering techniques were employed to explore alternative approaches to movie classification based on similarity in keyword associations. Different

linkage methods, such as complete, average, single, and centroid linkage, were tested to evaluate their effectiveness in identifying cohesive clusters of movies (Shetty & Singh 2021). Each linkage method resulted in a dendrogram representing the hierarchical relationships between movies based on keyword associations. We will examine the received topics and graphs in discussion section.

The ultimate goal of this classification process was to uncover underlying thematic patterns and associations among movies, providing insights into their content and facilitating further analysis and interpretation. Through these methods, movies with similar thematic elements or genres could be grouped together, offering valuable insights for various applications in film studies, recommendation systems, and audience segmentation.

Box-office Prediction

Data preparation

The initial step for data preparation involves visualizing the distribution of film revenues and other quantitative variables to detect the necessity of log-transformations. Subsequently, a prediction dataset is constructed, encompassing diverse attributes of cinematographic productions, such as film title, release month, funding source, genre, MPAA rating, duration, and various characteristics of the cast, production team, and country of origin. These variables are carefully processed and transformed into appropriate formats to facilitate data interpretation and analysis. For instance, categorical variables like release month and MPAA rating are converted into factor variables, while aggregate assessments of cast and production team attributes are computed to capture their overall contribution to the film.

The received frame is then split into training and testing subsets, ensuring a balanced representation of data across relevant thematic variables. Additionally, logarithmic transformations are applied to key quantitative variables, such as revenue and film duration, to address skewed distributions and improve adherence to standard modeling assumptions, to suppress the distance for variables with different distribution scales.

Machine learning models

For final assessment of box-office we used several machine learning models for the revenue of cinematographic productions forecasting (François, 2008). We begin by employing regression models, including Ordinary Least Squares (OLS), Decision Trees,

Random Forest, Boosting, and Stacking to capture the intricate relationships between various film attributes and their respective revenues.

For the OLS regression model, we fit the data using the lm() function in R, considering all available predictors. Subsequently, we evaluate the model's performance on both the training and testing datasets, assessing metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R2) values. Visualizations, including bar plots juxtaposing real, predicted, and production costs, provide intuitive insights into the model's efficacy.

Next, we explore decision tree-based approaches, visualizing the resulting decision trees to elucidate the model's decision-making process. Similarly, we evaluate the decision tree model's performance metrics on the training and testing datasets, facilitating comparative analysis with other regression techniques.

Furthermore, we employ ensemble methods such as Random Forest and Boosting to leverage the collective wisdom of multiple trees, enhancing predictive accuracy and robustness. We assess these models' performance and visualize their predictions alongside actual revenues and production costs.

Additionally, we investigate the efficacy of Stacking, a meta-ensemble technique that combines the predictions of multiple base models using a secondary meta-model. By aggregating diverse predictive insights, Stacking aims to achieve superior performance compared to individual models.

Lastly, we explore a neural network approach using TensorFlow in Python, building a multi-layer perceptron (MLP) regression model. We preprocess the data, including one-hot encoding categorical variables and standardizing numeric features. Subsequently, we train the neural network model and evaluate its performance on both the training and testing datasets, employing metrics such as MSE, RMSE, MAE, and R2.

Results

This section unveils the outcomes of our investigation into film revenue prediction. We present insights gleaned from semantic modeling, thematic distribution analysis, and machine learning techniques. By dissecting semantic structures, exploring thematic trends, and evaluating ML models, we illuminate the intricate dynamics shaping box office success.

Semantic Analysis

The analysis showed that the semantic model based on the NRC dictionary makes it possible to effectively assess the emotional states expressed in the text. Using the example of movie descriptions, we observed a variety of emotional reactions, from joy and expectation to fear and disappointment. The table of results (see Table 1) demonstrates the distribution of various emotions in the texts and the overall emotional coloring of each of them.

Table 1. Sentiment analysis results. Example (developed by the authors based on The Numbers data)

name	**sentiment**	**anger**	**anticipation**	**disgust**	**fear**	**joy**	**negative**	**positive**	**sadness**	**surprise**	**trust**
Avatar (2009)	2,85	9	5	3	7	3	11	12	5	4	6
Kung Fu Panda 4 (2024)	1,55	7	6	6	11	4	11	18	5	3	6
The Intouchables/1+1 (2011)	1,80	3	3	1	4	4	5	7	5	3	8
Sinister (2012)	-4,45	8	9	5	11	7	12	11	6	6	3
The Lobster (2015)	-2,90	4	0	1	5	2	6	6	4	0	2
NA	0,00	0	0	0	0	0	0	0	0	0	0

The conducted analysis utilizing the NRC lexicon enabled a detailed exploration of the emotional content within film descriptions. The results allow us to logically explain the ratings obtained for each text.

Avatar 2009

This film, depicting epic battles and romantic relationships, contains elements of action, romance, and fantasy. The text describes a struggle for survival on a mysterious planet inhabited by an alien species. Emotional ratings, such as fear (7) and negative sentiment (11), may be associated with tense battle scenes and conflict, while positive ratings (12) indicate the presence of romantic and inspiring moments in the film.

Kung Fu Panda 4 2024

Described as a family adventure, this film contains elements of action, comedy, and kindness. Emotional ratings, such as joy (4) and positivity (18), reflect comedic moments and positive aspects of the interaction between the main characters. High ratings for fear (11) may be associated with tense scenes and a sense of threat.

The Intouchables 2011

This film depicts an unexpected friendship between two individuals who initially appear vastly different. It contains elements of comedy, drama, and romance. High ratings for trust (8) and negative sentiment (5) may be associated with the theme of trust and the complex relationships between the characters. A low disgust rating (1) indicates the absence of negative emotions typically associated with this film.

Sinister 2011

This film describes a series of horrific murders and alarming discoveries in a family's home that was the victim of a crime. High ratings for fear (11) and negative sentiment (12) reflect tense and distressing moments present in the text. A low joy rating (7) and positive emotions (11) underscore the bleak atmosphere and alarming events of the film.

The Lobster 2015

This film depicts an alternative reality where people are subjected to strange rules and conditions for finding a partner. Ratings for various emotions, such as fear (5) and disgust (1), may be associated with the unusual and grim plot of the film. Low ratings for surprise (0) and trust (2) reflect the novelty and distrust of the rules of this world.

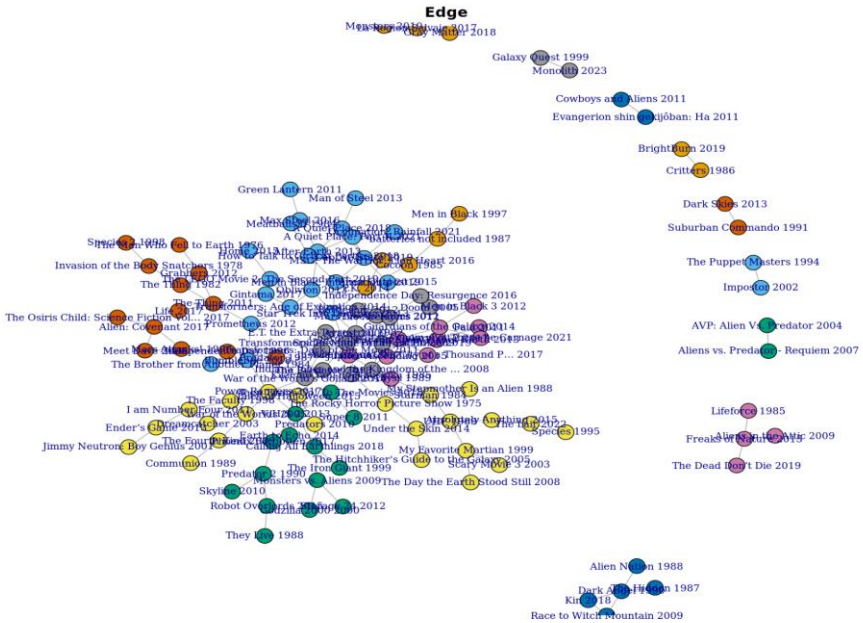
Thus, the analysis of emotional content within the texts allows for a logical interpretation of ratings, reflecting emotional nuances and the overall emotional tone of each film.

Topic Allocation

In the exploration of movie classification, a multifaceted approach was adopted to uncover thematic patterns embedded within synopses. Initially, the dataset underwent meticulous curation, focusing on movies containing the keyword "Alien" to streamline subsequent analyses. The ensuing steps involved parsing the synopses to extract keywords and constructing a bipartite graph representing movie-keyword associations.

Utilizing network analysis, diverse methodologies were employed to discern cohesive clusters within the constructed graph. Notably, community detection algorithms, including edge betweenness, fast-greedy, and walktrap, were applied to identify interconnected groups of movies based on shared thematic elements. These methods, detailed in Appendix 2, elucidate distinct strategies for partitioning the network into coherent communities. One of the methods can be seen in Figure 2.

Figure 2. Edge betweenness community detection (developed by the authors based on The Numbers data)



However, while networks effectively delineated thematic distributions on reduced datasets, their efficacy waned when confronted with the entirety of the data. Consequently, a strategic shift towards hierarchical clustering ensued, enabling robust analysis of larger datasets. Among the hierarchical clustering methods explored, complete linkage emerged as particularly effective, exhibiting superior performance in delineating logical groupings of movies based on textual descriptions. In contrast, alternative linkage methods led to

disparate groupings with varying movie counts, indicating their inadequacy in capturing cohesive thematic clusters. The received test trees can be seen in Appendix 3.

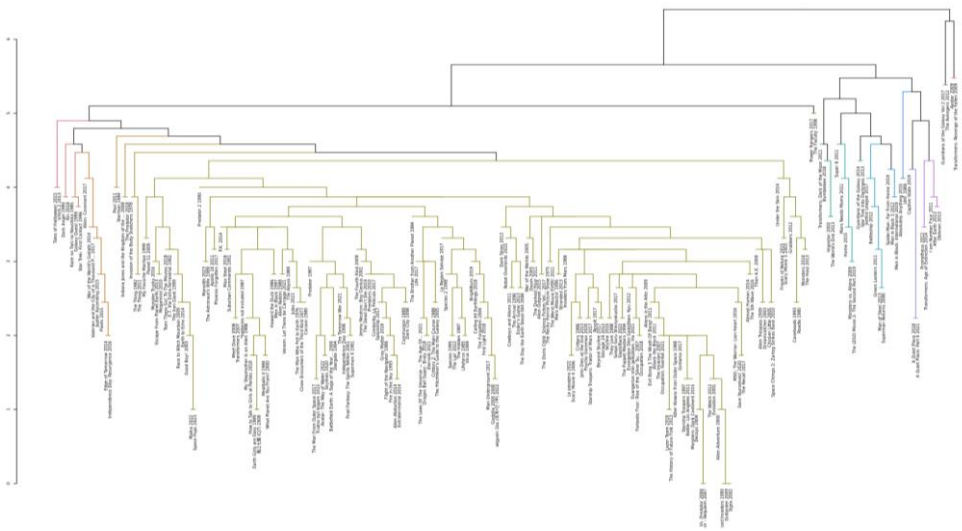
The hierarchical clustering techniques presented unique advantages and drawbacks, showcasing the diversity of approaches in analyzing textual data for topic classification. The brief overview can be found in Table 2.

Table 2. Hierarchical clustering algorithms difference

Method	**Distance Metric**	**Advantages**	**Disadvantages**
Complete Linkage	Maximum distance between any two points	Good for well-separated, spherical clusters	Sensitive to outliers, can create elongated clusters
Single Linkage	Minimum distance between any two points	Good for dense clusters with clear boundaries	Sensitive to chaining effects, can create long, irregular clusters
Average Linkage	Average distance between all pairs of points	Good balance, less sensitive to outliers and chaining effects	N/A
Centroid Linkage	Distance between cluster centroids	Computationally efficient	Assumes spherical clusters, may not be suitable for irregular shapes

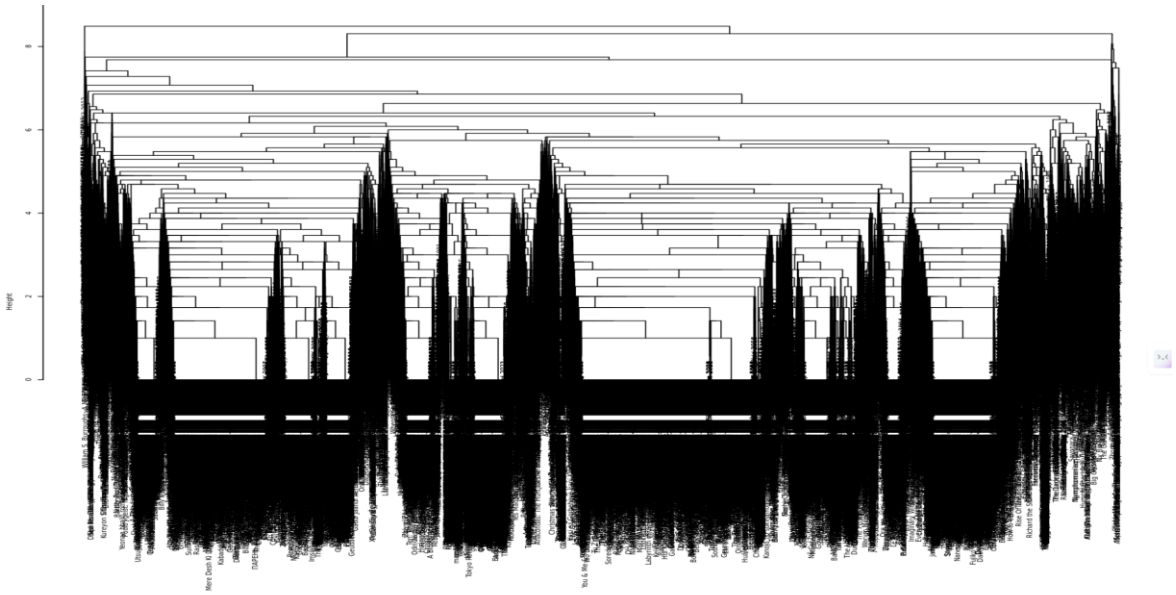
Notably, complete linkage demonstrated proficiency in revealing coherent thematic clusters, underscoring its utility in uncovering nuanced patterns within movie synopses. The hierarchy on test group of films can be seen in Figure 3.

Figure 3. Hierarchical clustering with complete linkage method (developed by the authors based on The Numbers data)



After applying the selected hierarchical clustering method to the complete data set, the results are shown in Figure 4. For further analysis and use, it was necessary for the optimal level of tree trimming, which would ensure an acceptable distribution of films into groups, preserving thematic diversity and providing coverage of a larger amount of data.

Figure 4. The full hierarchical tree (developed by the authors based on The Numbers data)



For this purpose, an iteration algorithm was written in which the iterative application of the tree pruning method with different height values was carried out. At each iteration statistics were calculated, including the number of topics detected, the total number of films in groups, and the percentage of films covered by these groups. Ultimately, based on the data obtained, it was decided to choose the optimal level of tree pruning. The full assessment can be observed in Appendix 4.

The most attractive result was achieved when the height of the tree cut was 4 resulted in 30 topics, covering approximately 81.27% of the films. At this level, the optimal number of topics was found, while the percentage of films covered by these topics made up a significant part of the entire data set. Additionally, the distribution of films within large groups turned out to be approximately uniform, which underlines the effectiveness of the chosen approach in ensuring equilibrium coverage of thematic groups.

Econometric Estimations

We employed regression analysis to analyze two models: model 1, based on real values, and model 2, based on logarithmic values. In Model 1, where real values are used as the dependent variable, the R-squared value is 0.617, indicating that approximately 61.7% of the differences in box office receipts worldwide can be explained by independent variables.

The adjusted R-square, taking into account the number of predictors, is 0.607. The F-statistic, equal to 62.030, is significant at $p < 0,01$, which suggests that the model fits well. The following independent variables resulted in being of high significance at the $p<0,01$ level in Model 1: different months, Source based on a musical or opera, Genre, Running Time, Lead Power, Support Power, Producer Power, N_countries, Prod_country_Power, MPAA rating and is_franchise1.

In Model 2, which uses logarithmic values for the variables, the R-squared value is 0.605, indicating that approximately 60.5% of the variance in the global box office logarithm can be explained by independent variables. The adjusted R-square is 0.595. The F index of 58.920 is significant at $p < 0.01$, which indicates a good fit of the model. The independent variables that resulted in being of high significance at the $p<0,01$ level in Model 2 are months, Genre, Running Time, Lead Power, Support Power, Producer Power, N_countries, Prod_country_Power, MPAA rating and is_franchise1.

Comparing the two models, it can be noted that model 1 has a slightly higher R-squared value, which suggests that it explains a slightly larger proportion of the variance of the dependent variable compared to model 2. However, both models demonstrate similar patterns of significance of independent variables, which indicates a stable relationship between predictors and global box office receipts in the models. Both models exhibit significant explanatory power, as indicated by their respective R-squared values. The coefficients of various independent variables provide insights into their impact on the worldwide box office, with different magnitudes between the two models due to the transformation of the dependent variable. As for the sentiment estimates, such variables, as disgust, joy, sadness as surprise seem to have a significant impact on the film financial performance in both models, while fear, trust – in model 1, and anger – in model 2. The full table with the obtained results is demonstrated in the Appendix 5.

Machine Learning Prediction

Random forest emerged as the superior model based on the provided training and testing data results. The full results can be found in Table 3 – Table 6, which examines the error terms of the models in predicting the final value of film box-office. The visualization of the obtained results can be found in Appendix 6.

Table 3. Training data results

Model	MSE	RMSE	MAE	R^2
OLS	2.050 * 10^6	143,182,468.2128	85,948,836.4372	0.6183
Tree	2.089 * 10^6	144,529,459.026	83,524,726.858	0.611
RandomTree	$4.108 * 10^{15}$	64,091,542.9528	32,926,450.5441	0.9235
Boosting	$4.122 * 10^{15}$	64,203,467.5616	43,302,483.6853	0.9232
Stacking	$1.623 * 10^{15}$	40,289,108.1540	26,772,583.2468	0.9698

Table 4. Testing data results

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Stacking	$1.623 * 10^{15}$	40,289,108.1540	26,772,583.2468	0.9698

Table 5. Training log-data results

Model	MSE	RMSE	MAE	R^2
OLS	1.8180	1.3483	0.9559	0.6082
Tree	1.9598	1.3999	0.9893	0.5777
RandomTree	0.3105	0.5572	0.3724	0.9331
Boosting	0.6776	0.8232	0.6086	0.8540
Stacking	0.1244	0.3528	0.2522	0.9732

Table 6. Testing log-data results

Model	MSE	RMSE	MAE	R^2
OLS	1.7450	1.3210	0.9589	0.5752
Tree	2.1709	1.4734	1.0037	0.4715
RandomTree	1.5032	1.2261	0.8388	0.6340
Boosting	1.6180	1.2720	0.8782	0.6061
Stacking	1.5382	1.2402	0.8569	0.6255

Random forest achieved the lowest MSE and RMSE values across both training and testing datasets compared to other models. This indicates that random forest's predictions were closest to the actual values, suggesting better overall performance in terms of minimizing prediction errors. Similarly, it exhibited the lowest MAE values, implying that its predictions had the smallest average absolute deviations from the actual values. This indicates robustness and accuracy in predicting movie box office performance.

Random forest also demonstrated the highest R^2 values, indicating the proportion of variance in the dependent variable (movie box-office revenue) that is predictable from the independent variables included in the model. Higher R^2 values suggest that random forest captured more variability in the data and provided better predictions compared to other models.

Considering these metrics collectively, and taking into account the minimum level of potential overfitting in comparison with other machine learning methods, random forest outperformed other models in terms of accuracy, error terms, and overall predictive power. Its ability to handle complex relationships, nonlinearities, and interactions among predictor variables likely contributed to its superior performance. The results of random forest prediction can be examined in Figures 5 and 6.

Figure 5. Box-office prediction on testing data (developed by the authors based on The Numbers data)

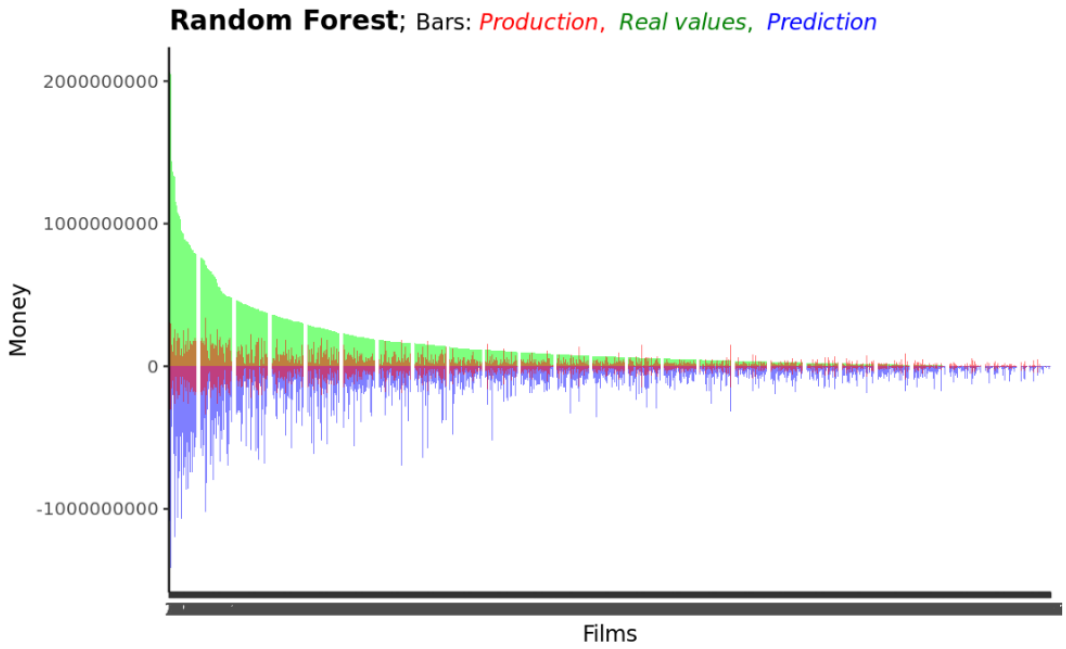
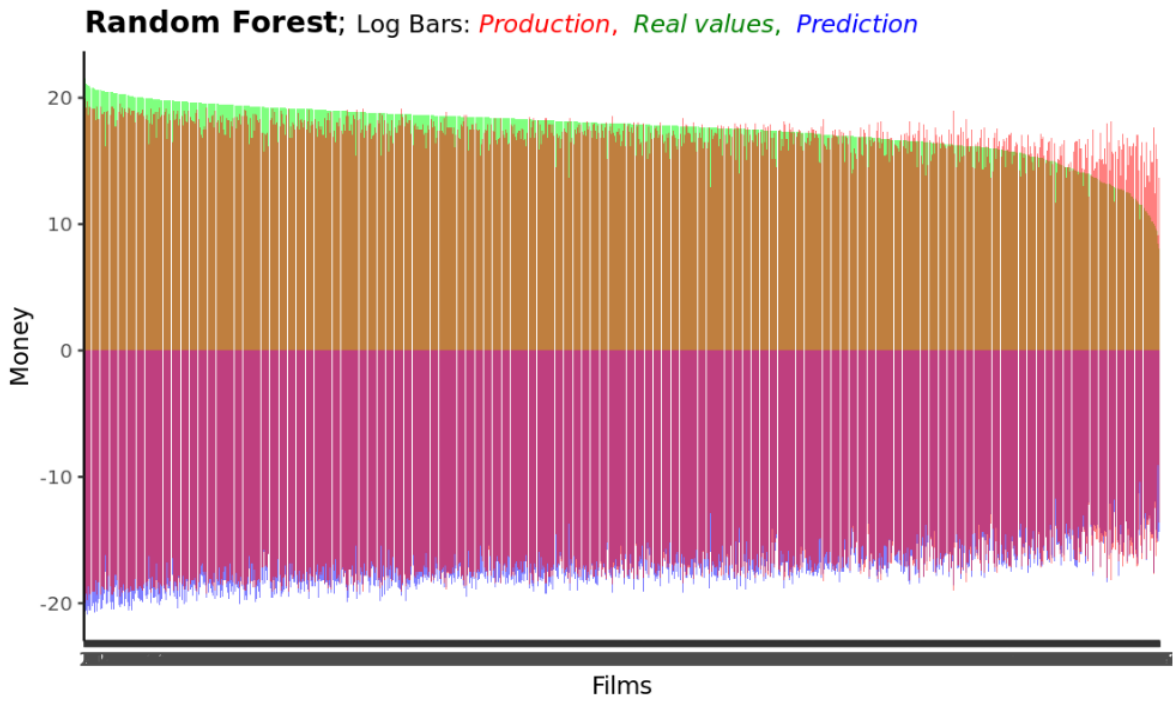
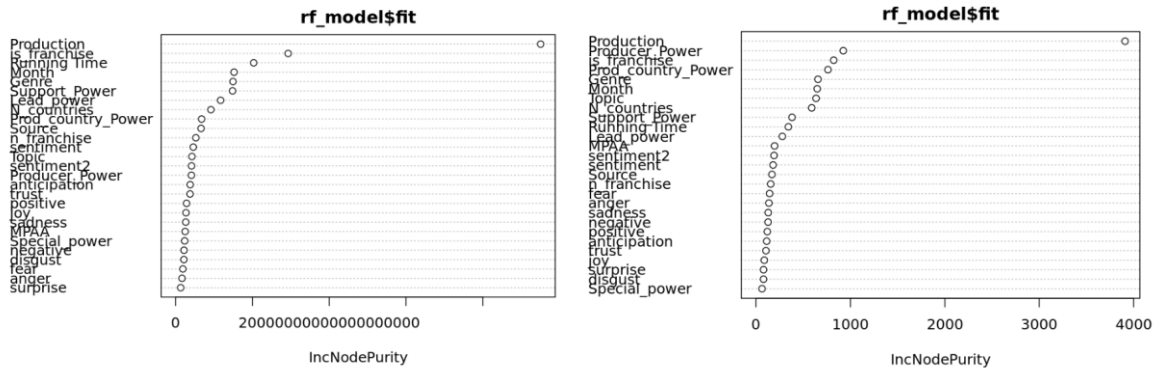


Figure 6. Box-office prediction on testing log-data (developed by the authors based on The Numbers data)



Due to the transparency of the random forest mode construction, we are able to assess the importance of variables that are considered by our trees in their predictions. The variable "IncNodePurity" is a measure used in random forest models to evaluate the importance of each predictor variable in the model's decision-making process. It represents the decrease in impurity achieved by splitting the data based on a particular predictor variable. Higher values of IncNodePurity indicate that the variable contributes more to reducing impurity and, therefore, is considered more important in the model. The visual representation of important variables can be observed in Figure 7.

Figure 7. Variable importance graphs (real & log) (developed by the authors based on The Numbers data)



A brief explanation of the important variables based on their IncNodePurity values in both log values and real values:

- Production variable, both in log values and real values, has a high IncNodePurity, indicating its significant contribution to reducing impurity in the decision-making process of the random forest model. It likely represents the production budget of the movie, suggesting that higher-budget productions tend to have a greater impact on predicting the outcome variable (e.g., movie box office revenue).
- Month: The month of release of the movie also shows a notable IncNodePurity, implying that it plays a crucial role in determining the movie's success. Seasonality or timing of release could influence audience behavior and, consequently, movie revenue.
- Source variable represents the source material of the movie, such as books, plays, or original screenplays. The IncNodePurity suggests that the source material significantly influences the movie's success, with certain sources having a more substantial impact than others.
- Genre: Different movie genres exhibit varying degrees of importance, as indicated by their IncNodePurity values. Certain genres may have a more significant effect on audience preferences and box office performance.
- MPAA: The movie's rating by the Motion Picture Association of America also appears to be an important predictor, suggesting that audience age restrictions or content advisories influence movie revenue.
- The length of the movie plays a role in its success, with longer or shorter films potentially affecting audience engagement and box office performance.
- Lead_power variable represents the star power or popularity of the lead actor/actress in the movie and indicate that casting decisions can impact a film's commercial success.
- Whether the movie is part of an existing franchise seems to be a significant predictor, with franchise movies potentially enjoying built-in fan bases and higher box office returns.

These variables, with their high IncNodePurity values, provide valuable insights into the factors that contribute most to predicting movie box office performance within the context of the random forest model.

Neural Network Prediction

Despite the high machine accuracy of our model, we decided to test how a simple multiplayer perceptron would cope with the task of predicting a box-office movie. We constructed a neural network model for regression tasks using the Keras library, Python. The architecture of the neural network consisted of four fully connected layers, with ReLU activation functions applied to the hidden layers and a linear activation function for the output layer. The sequential model was compiled using the Adam optimizer with a learning rate of 0.001 and mean squared error (MSE) loss function. The model was trained for 100 epochs with a batch size of 32. The predicted values of our neural network can be analyzed in Figures 8 and 9.

Figure 8. Fitting predicted and real values on train and test data (developed by the authors based on The Numbers data)

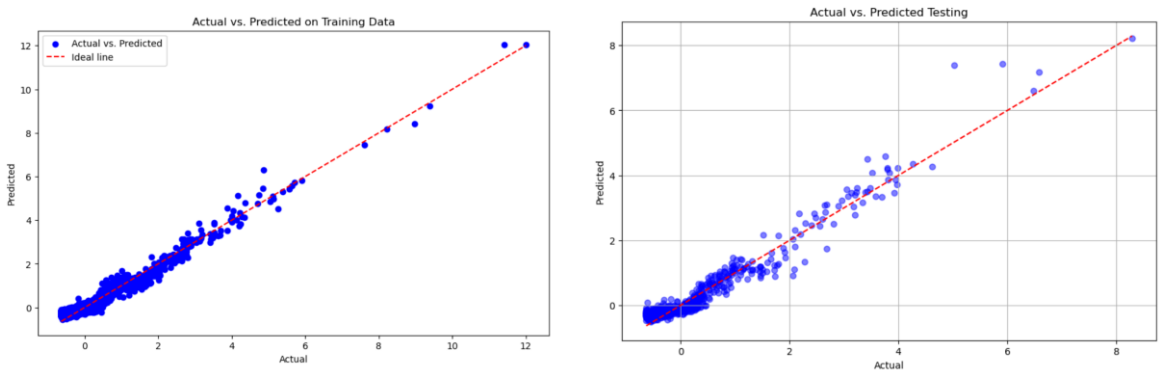
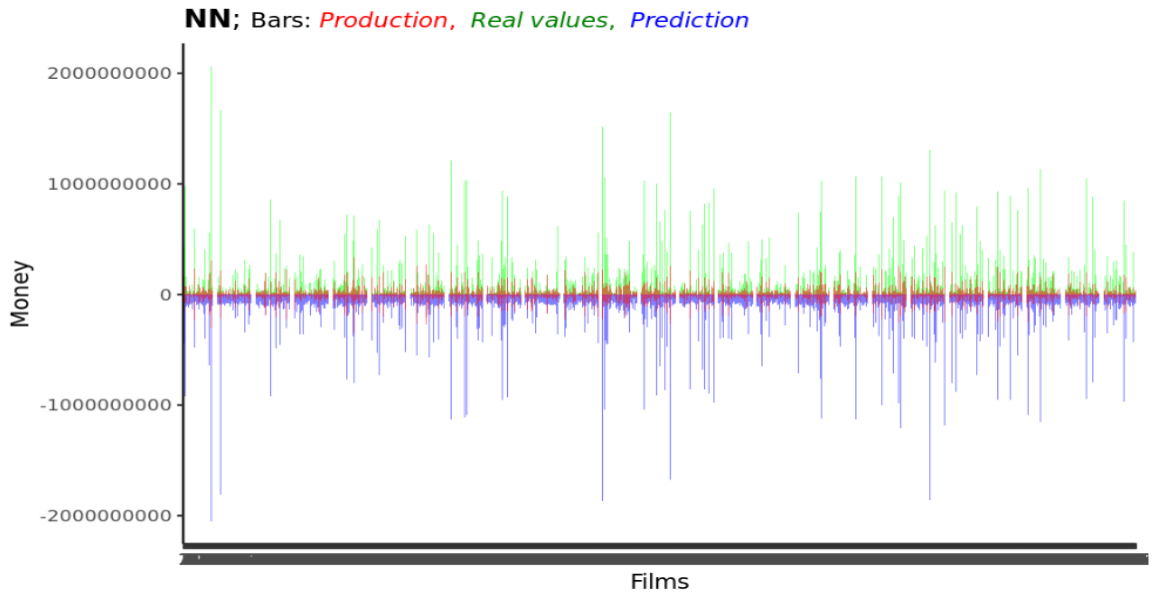


Figure 9. Box-office prediction on testing data (developed by the authors based on The Numbers data)



The performance metrics calculated in Table 7 indicate that the neural network model achieved high accuracy in predicting the target variable on both the training and testing datasets, as evidenced by the low values of MSE, RMSE, and MAE, as well as the high R-squared values compared to the previously used other machine learning algorithms. For the analysis of managerial accuracy of our models we will use both random forest and neural network algorithms, because one of them is the leader of simple prediction models meanwhile is interpretable, while another is a general winner.

Table 7. Neural network prediction evaluation

Metric	Train	Test
MSE	0,061	0,075
RMSE	0,247	0,274
MAE	0,209	0,218
R2	0,937	0,93

Finally, referring to the factors, that are most worthy of being considered while making a decision on which movies to invest into, and based on the analysis conducted above (econometrics estimations, results of random forest "IncNodePurity" variable prediction), the following can be concluded.

The first hypothesis states: Content Factors (Genre, Topic, Source, Franchise, Emotions) have a statistically significant effect on the financial performance of a movie at the box office. We can accept this hypothesis. More specifically, adventure, horror, and musical genres of a movie, folk fairytale, legend, musical, opera, and play as a source for scenario of a movie, film being a part of a franchise, and such emotions, as joy, sadness, anger, disgust, fear, surprise, and trust (however, with the surpass of positive emotions) are the content factors having the most significant impact on the financial performance of a movie.

The second hypothesis states: Production Factors (Production Method, Film Time, Acting Power, Directing Power, and Producer Power) have a statistically significant effect on the financial performance of a movie at the box office. This hypothesis can be accepted. More precisely, the method of film production, the run time of a movie, the power factor of lead and support actors with lead actors prevailing in importance, as well as directing and producing power indeed demonstrated to be significant factors, signaling the possible future success at the box office.

The third hypothesis states: Commercial Factors (Seasonality by month, Budget, MPAA rating, Release Countries (number)) have a statistically significant effect on the

financial performance of a movie at the box office. This hypothesis can be accepted. The budget factor, the number of countries where the movie is released, as well as its country of origin, MPAA rating, and seasonality of release by month, especially the release time during high seasons – May, June, July, and December, showed the most influence on a project's financial success.

Discussion

In evaluating the accuracy of our predictive models, particularly the random forest and neural network models, several key insights emerge. Firstly, from a managerial perspective, both models demonstrate commendable accuracy, as evidenced by their respective accuracy scores of 78.51%. However, upon closer examination, notable differences in performance metrics become apparent.

Investment Opportunities

In the Discussion section, we have opted to conduct an analysis of managerial effectiveness regarding our decision-making models.

To assess the quality of our predictive models, we propose envisioning an ideal model that accurately predicts, with 100% precision, the films in which investment is warranted, and furthermore, accurately estimates the potential revenue. In other words, our hypothetical model predicts a film with $100,000.00 expenditure to generate $2,000,000.00 in revenue, and in reality, the film indeed earns this amount in the future. Thus, a hypothetical investor, based on the models, can make decisions about whether or not to invest in a film, under the condition that they are guaranteed to receive all the film's earnings if they simply cover the film's costs in full. Suppose our investors follow a strategy wherein if the film breaks even according to the model's estimates, they invest in it.

Consequently, we will not only obtain machine accuracy of the model, but also examine managerial accuracy, which evaluates the financial losses incurred by investors when using our models.

Random forest results

From a managerial perspective, the results of our random forest model indicate several key insights, which can be observed from the Table 8.

Table 8. Confusion matrix of investment-friendly films based on random forest

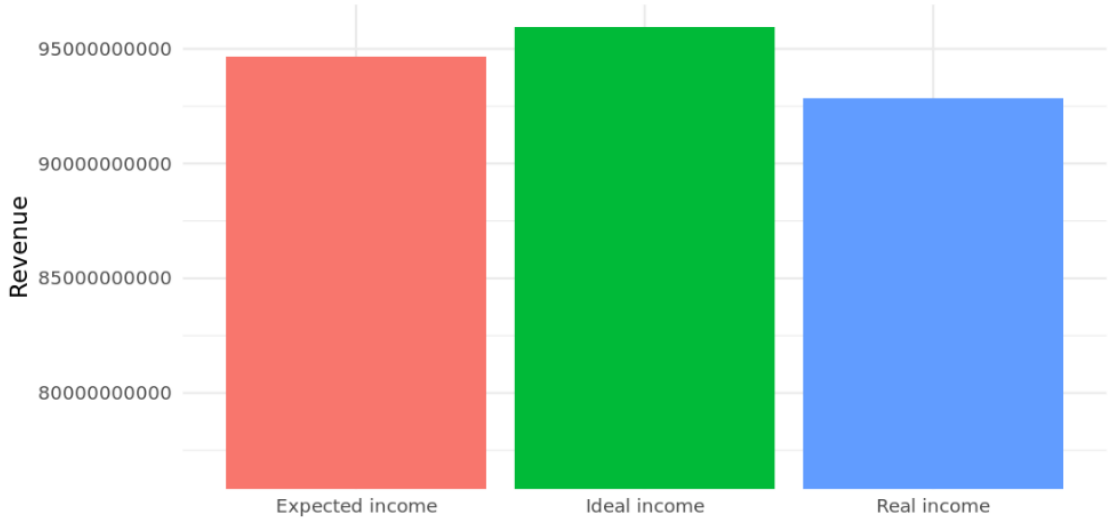
Prediction|Truth	FALSE	TRUE
FALSE	4	1
TRUE	203	618

Firstly, examining the confusion matrix, we observe that the model's predictions align with the truth in the majority of cases, as evidenced by the relatively high accuracy of 78.51%. However, the sensitivity (true positive rate) is notably low at 5.53%, indicating that

the model has difficulty correctly identifying positive cases, which in this context represent films that would generate significant revenue. On the other hand, the specificity (true negative rate) is high at 99.56%, suggesting that the model performs well in identifying negative cases, i.e., films that would not generate significant revenue.

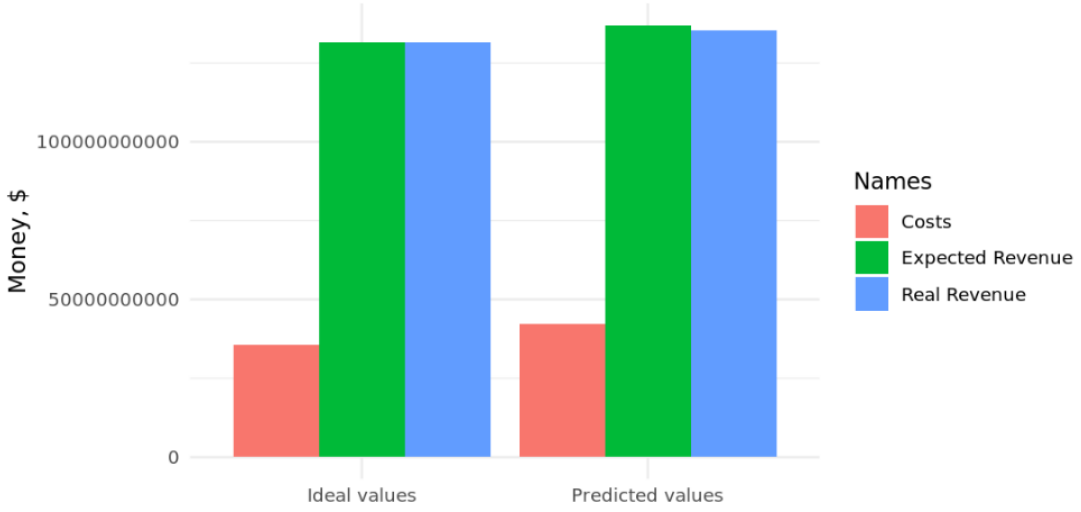
Furthermore, assessing the financial implications described in Figure 10, we compare the ideal, expected, and real incomes derived from the model predictions. The difference between the expected and ideal incomes is -$1,322,087,454, indicating a slight underperformance of 1.38%. Similarly, the difference between the expected and ideal incomes is -$1,799,911,372, representing a larger deviation of 1.9%. Notably, the difference between the real and ideal incomes is -$3,121,998,826, indicating a more significant deviation of 3.25%.

Figure 10. Investors' income comparison based on random forest (developed by the authors based on The Numbers data)



Delving deeper into the financial analysis graphically represented in Figure 11, we compare the predicted and ideal values for costs, expected revenue, and real revenue. The predicted revenue exceeds the ideal revenue by $5,491,021,546 or 4.17%, suggesting an optimistic projection by the model. However, the real revenue falls short of the ideal revenue by $3,691,110,174 or 2.81%, indicating a more conservative estimate. Moreover, the real costs exceed the ideal costs by $6,813,109,000 or 19.16%, highlighting potential inefficiencies in cost management.

Figure 11. Financial estimations based on random forest (developed by the authors based on The Numbers data)



Overall, while the random forest model demonstrates respectable accuracy, there are notable discrepancies between predicted and actual financial outcomes, indicating the need for further refinement and consideration of managerial implications in decision-making processes.

Neural network results

From the managerial perspective, the neural network model demonstrates moderate accuracy in predicting investment decisions stated in Table 9, as evidenced by an accuracy of 78.51%. However, the kappa statistic indicates only slight agreement beyond chance (7.61%), suggesting that the performance of the model is not significantly better than random guessing. The sensitivity (5.53%) highlights the model's weakness in correctly identifying positive cases (investment-worthy films), while the specificity (99.57%) indicates its strong performance in recognizing negative cases (films not worthy of investment).

Table 9. Confusion matrix of investment-friendly films based on neural network

Prediction|Truth	FALSE	TRUE
FALSE	11	3
TRUE	188	687

Firstly, as we did it previously, we compare the ideal, expected, and real incomes generated from our film investments, visualized in Figure 12. The ideal income, representing an optimistic projection of revenue generation, amounts to $100,779,213,025.00. In contrast, the expected income, based on our predictive models, is higher at $120,679,128,571.00,

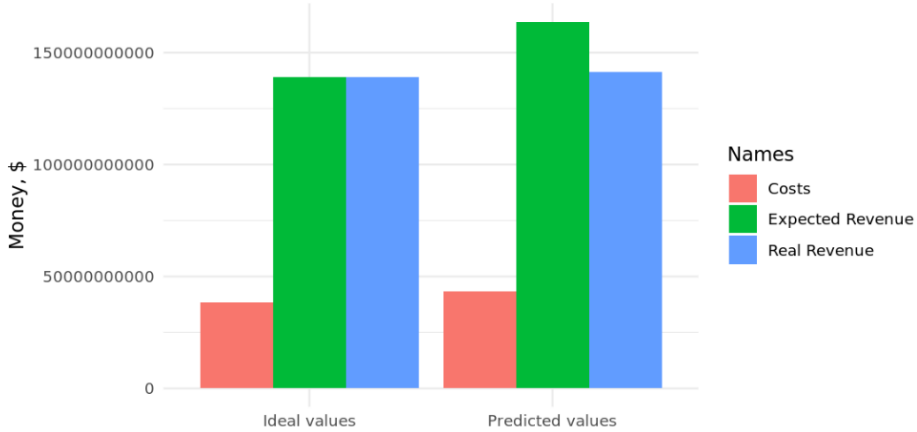
reflecting a more optimistic outlook on revenue prospects. However, the real income falls short of both the ideal and expected values, amounting to $98,377,388,811.00. This disparity highlights the challenges and uncertainties inherent in forecasting financial outcomes in the film industry.

Figure 12. Investors' income comparison based on neural network (developed by the authors based on The Numbers data)

The differences between the expected-ideal, expected-real, and real-ideal incomes further elucidate the discrepancies and risks associated with relying solely on predictive models for investment decisions. The expected-ideal difference of $19,899,915,546.00 or 19.75% underscores the optimism in revenue projections, while the negative expected-real difference of $-22,301,739,760.00 or -18.48% and the negative real-ideal difference of $-2,401,824,214.00 or -2.38% emphasize the potential deviations from projected revenues and the associated financial implications.

Next, we analyzed the costs and revenues associated with our investment decisions to gain a comprehensive understanding of their financial impact. The full financials can be found in Figure 13.

Figure 13. Financial estimations based on neural network (developed by the authors based on The Numbers data)



The ideal costs, representing the projected production expenses, amount to $38,302,259,000.00. Correspondingly, the ideal values for expected and real revenues stand at $139,081,472,025.00, reflecting the anticipated revenue generation under ideal cost conditions.

In contrast, the predicted costs based on our models amount to $43,142,444,000.00, indicating potential cost overruns or deviations from ideal projections. Despite these deviations, the predicted expected revenue exceeds the ideal revenue by $24,740,100,546.00 or 17.79%, suggesting an optimistic outlook on investment returns. However, the comparison between real and ideal revenues reveals a more conservative picture, with the real revenues lagging behind the ideal values by $2,438,360,786.00 or 1.75%.

Furthermore, the comparison between real and ideal costs underscores the potential impact of cost overruns on investment profitability. The real costs exceed the ideal costs by $4,840,185,000.00 or 12.64%, indicating the challenges in managing production expenses effectively.

The financial analysis highlights the complexities and uncertainties involved in film investment decisions. While predictive models offer valuable insights into revenue prospects, their limitations in accurately forecasting financial outcomes underscore the importance of exercising caution and complementing quantitative analysis with qualitative judgment and industry expertise. Thus, the need for an integrated approach to investment decision-making, taking into account both financial forecasts and operational realities in the film industry, is emphasized by the discrepancies between ideal, expected and real revenues, as well as the associated differences in costs.

Comparing and contrasting with the prior scientific literature

The emphasis of the paper's methodology on internal signals such as star power, production budgets, genre, and storyline is strongly supported by Fan et al. (2021), who also identify these elements as critical quality indicators that influence audience expectations and box-office performance. Both studies highlight how a high-profile cast and substantial budget can signal high quality to consumers, thereby attracting more viewers. While Fan et al. (2021) and other researchers recognize the importance of internal signals, they also integrate external factors such as electronic word-of-mouth (eWOM) and reviews. On the one hand, the research by Hofmann et al. (2017) and Fan et al. (2021) on the influence of star power and eWOM further validates our findings. These studies show that star power significantly affects box-office receipts, particularly during the premiere week, which aligns

with our conclusion that high-profile actors and substantial production budgets are crucial internal signals. On the other hand, with our focus on the internal characteristics, certain essential relationships could be missed in the results we obtained, that were considered by other researchers.

The integration of neural networks and machine learning models, as discussed by Sharda and Delen (2006) and Delen et al. (2007), further complements our focus on internal signals. These studies demonstrate that advanced algorithms, when fed with relevant internal data like production budget and star power, can achieve high predictive accuracy. Our paper supports this by showing how internal signals effectively forecast financial outcomes, particularly in the early stages of production when external signals are less measurable. However, apart from the researchers above, we give equal weight to the potential factors, while the researchers specifically highlight star power and production budgets.

Our paper also intersects with the sentiment analysis research by Hur et al. (2016) and the social media analysis by Gaikar & Marakarkandy (2015). These studies highlight the impact of audience-related factors such as pre-release sentiment and electronic word-of-mouth (eWOM) on box-office forecasting. While our primary focus was on internal signals, the external factors described in the papers mentioned above offer complementary insights post-release. This suggests that a holistic approach, incorporating both internal and external signals, can enhance predictive accuracy across a movie's lifecycle.

Moreover, the study by Ghiassi et al. (2015) introducing novel variables like production budget and pre-release advertising expenditures reflects our findings. Both studies confirm that these early-stage factors are pivotal in shaping financial forecasts. However, apart from this study, that leverages machine learning and hybrid models to predict box-office success, our research adopts a more traditional analytical framework focused on internal factors.

Nevertheless, the specific context of our study may differ from others, that is also an important point to discuss. For example, Gaikar & Marakarkandy (2015) and Zhang (2022) emphasize the role of social media sentiment in different cultural contexts (Bollywood and Chinese film industry, respectively). Our study, by focusing on internal signals applicable across various contexts, offers more universally applicable insights, though it may lack the cultural specificity that can be crucial in some markets.

Conclusion

The bachelor thesis is aimed to determine the early-stage production factors that significantly influence a film's financial success at the box office and to develop a predictive tool to aid investment decisions. The comprehensive analysis, which included semantic modeling, thematic distribution analysis, machine learning techniques, and econometrics analysis has yielded valuable insights aligned with the research question and theoretical framework of the current paper.

Firstly, the semantic analysis using the NRC lexicon provided a detailed exploration of the emotional content within film descriptions. This analysis revealed that emotional factors such as fear, joy, and trust play a crucial role in shaping audience reactions and, consequently, box office performance. For instance, films like "Avatar" and "Sinister" showed high ratings for fear and negative sentiments, which correlated with their dramatic and intense storylines. Conversely, "Kung Fu Panda 4" exhibited high ratings for joy and positive sentiment, reflecting its family-friendly and comedic nature. These findings support the idea that emotional estimators, among other content-related signals, are an essential driver of audience interest and further financial success of the project.

Secondly, the thematic analysis, employing advanced clustering methods such as community detection algorithms and hierarchical clustering, uncovered distinct thematic patterns within movie synopses. These methods allowed us to group movies based on shared thematic elements, revealing how thematic coherence can influence box office outcomes. For example, the hierarchical clustering identified that genres like adventure, horror, and musicals consistently formed coherent thematic clusters that performed well financially. This supports the notion that thematic consistency and genre-specific appeal, among other content-related signals, are critical factors in predicting a film's success.

Moreover, econometric estimations provided two robust models: one based on real values and the other on logarithmic values. In Model 1 (real values), the R-squared value was 0.617, indicating that 61.7% of the variance in box office receipts is explanatory by the independent variables. Significant predictors included production method, genre, source material, running time, star power (lead and support), and seasonal release timing. Model 2 (logarithmic values) showed an R-squared value of 0.605, with similar significant predictors. Both models highlighted the stable relationship between these factors and global box office revenues, emphasizing the theoretical premise that production quality and strategic release timing, among other production-related signals, are pivotal to a film's financial success.

Furthermore, considering the factors of accuracy and predictive power, random forest is shown to be the most effective model among other employed. The model's high R-squared values and low error terms underscored its reliability. Key variables such as production budget, release month, source material, genre, MPAA rating, and franchise status emerged as significant predictors. For instance, high-budget productions, releases during peak months (May, June, July, December), and films based on well-known sources or franchises were consistently associated with higher box office revenues. The random forest model's transparency allowed us to assess the significance of these variables, further validating their impact on financial outcomes.

Additionally, our neural network model confirmed the robustness of these findings, providing further validation. The neural network achieved high accuracy, with metrics indicating strong predictive capabilities for both training and testing datasets.

From a theoretical perspective, the paper contributes to the understanding of how emotional engagement, thematic coherence, and strategic production decisions influence box office success. By integrating emotional content and thematic analysis into predictive models, we offer a novel approach to understanding audience preferences and film success.

From a managerial standpoint, these findings offer actionable insights for stakeholders across the film production landscape. Filmmakers stand to benefit from the ability to assess the financial potential of projects during the early stages of development, while investors can make more informed decisions regarding film financing based on the model's predictions. Ultimately, this research holds the promise of contributing to a more data-driven and financially sound film production ecosystem, empowering stakeholders to navigate the complexities of the industry with confidence.

Limitations and Future Research Agenda

However, to enhance the research's robustness and generalizability, we should state some limitations and considerations for improvement:

Data Considerations

While "The Numbers" offers a valuable source of box office data, a more holistic understanding of a film's financial success can be achieved by incorporating additional metrics. This includes production costs, profit margins, and revenue streams beyond box office collections, such as streaming services and merchandise sales. Data from these additional sources can be integrated alongside the box office information to provide a more comprehensive picture of a film's financial performance.

However, the quality of the data is paramount. Web scraping inherently introduces errors, so implementing robust data cleaning and validation techniques is crucial to ensure the accuracy of our analysis.

Furthermore, the research currently focuses on historical data. To strengthen the model's predictive power, we should explore methods to integrate external factors that might influence future box office success. This could involve incorporating data on economic trends or evolving audience preferences. By considering these external factors, the model can move beyond simply analyzing past performance and offer insights into a film's potential future financial viability.

Algorithm Considerations

Machine learning models are powerful tools for pattern recognition and can be instrumental in identifying the factors influencing a film's financial success. However, it's important to acknowledge the inherent uncertainty in future predictions. Additionally, depending on the chosen algorithms, the model might become a "black box," making it difficult to understand the specific reasons behind its predictions. To address this, utilizing interpretable machine learning models could provide valuable insights. These models offer greater transparency into the factors driving the model's predictions, allowing filmmakers and investors to make more informed decisions.

Future Research Agenda

By expanding the scope of analysis, a fuller understanding of a film's success can be achieved, despite an initial focus on early production data being a valuable insight. This includes incorporating factors that influence success throughout the entire production cycle.

By analyzing elements like directorial execution, marketing campaign effectiveness, and critical reception alongside early production data, we can gain a more holistic picture of what drives a film's financial performance. Directorial execution can significantly impact audience engagement, while marketing campaigns play a crucial role in generating awareness and interest. Critical reception, both positive and negative, can influence a film's box office performance and ancillary revenue streams.

Moreover, quantitative data, while valuable, can offer a limited perspective on audience preferences. Supplementing this data with qualitative analysis can enrich the model's understanding of these preferences. Techniques like sentiment analysis of social media reviews can reveal audience reception to specific themes, genres, and overall tone.

In conclusion, our research underscores the significant impact of content, production, and commercial factors on film financial performance, providing a comprehensive understanding of the box office success components and offering practical applications for industry stakeholders. In response to the limitations of the data and incorporating the aforementioned considerations, the future research can achieve a more robust and nuanced understanding of financial performance prediction in the film industry.

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Appendix 1

Table 1. *Summary of the discussed articles*

Authors and journal	Research question	Method and data	Measurements (IV/DV)	Main findings
Ahmed, U., Waqas, H., & Afzal, M.T. (2020). Pre-production box-office success quotient forecasting. Soft Computing, 24, 6635–6653.	How can different features be utilized to predict the success of a movie in the early stages of development?	Data scraped from IMDb and social media platforms Supervised machine learning, gradient boosting, extreme gradient boosting, random forest, support vector machine	Independent variables: Director and cast Facebook likes, IMDb ratings, movie genres. Dependent variable: Movie success classification	The hybrid machine learning model achieved an 85% accuracy rate in predicting movie success during the pre-production phase, leveraging features such as director and cast popularity on social media and IMDb ratings.
Delen, D., Sharda, R., & Kumar, P. (2007). Movie forecast Guru: A Web-based DSS for Hollywood managers. Decision Support Systems, 43(4), 1151-1170.	How can decision support systems be utilized to predict the box-office success of movies in the film industry?	Neural Networks, Decision Trees, Ordinal Logistic Regression, and Discriminant analysis, information fusion techniques; dataset of 849 movies from ShowBiz Inc., Internet Movie Database (IMDb)	Independent variables: MPAA rating, competition level, star value, genre, special effects, sequel status, and number of screens. The dependent variable is box-office gross revenues	The creation of WEB-based DSS tool what can be used by Hollywood managers
Sharda, R., & Delen, D. (2006). Predicting box-office success of motion pictures with neural networks. Expert Systems with Application	Predicting the box-office success of motion pictures using neural networks.	Data on 834 movies released between 1998 and 2002, sourced from ShowBiz Data, Inc. The method involves using stratified 10-fold cross-validation to estimate the performance of neural network classifiers.	Independent variables: MPAA rating, competition level, star value, content category, technical effects, sequel status, and the number of	The neural network model demonstrated superior predictive accuracy compared to traditional statistical methods, achieving higher classification accuracy in categorizing movies into different box-office success categories with pinpoint accuracy

s, 30, 243–254.			screens. The dependent variable is the box-office success category of movies.	36.9% and 75.2% within a category. Key determinants of box-office performance, including the number of screens, high technical effects, and star value.
Ghiassi, M., Lio, D., & Moon, B. (2015). Pre-production forecasting of movie revenues with a dynamic artificial neural network. Expert Systems with Applications, 42(6), 3176-3193.	How does the DAN2 model compare to the other models in box-office revenue forecasting and classification?	DAN2 forecasting and classification, initialized with KNN, and optimized using grid-search.	Independent variables: Production budget, pre-release advertising expenditures; Dependent variables: Box-office revenue forecasts, classification accuracy.	DAN2 outperforms the models proposed by Sharda and Delen (2010), achieving higher accuracy in box-office revenue forecasting and classification. KNN initialization and grid-search optimization contribute to DAN2's effectiveness. Inclusion of novel variables enhances predictive capability, while the limited impact of advertising expenditures suggests their minor role in forecasting outcomes.
Delen, D., & Sharda, R. (2010). Predicting the financial success of hollywood movies using an information fusion approach Industrial Engineering Journal, 21(1), 30–37.	Predicting the financial success of Hollywood movies using an information fusion approach.	Decision Trees, Support Vector Machines, Ensemble Methods. Dataset of 2632 movies released between 1998 and 2006	Independent Variables: various factors related to movie production, Distribution and marketing efforts. Dependent Variable: Box-office gross revenues.	The ensemble method, particularly information fusion, outperformed individual prediction models. SVM exhibited the best performance among individual models.
Qiu, Y., &	How can	Regularization-	Independent	The study

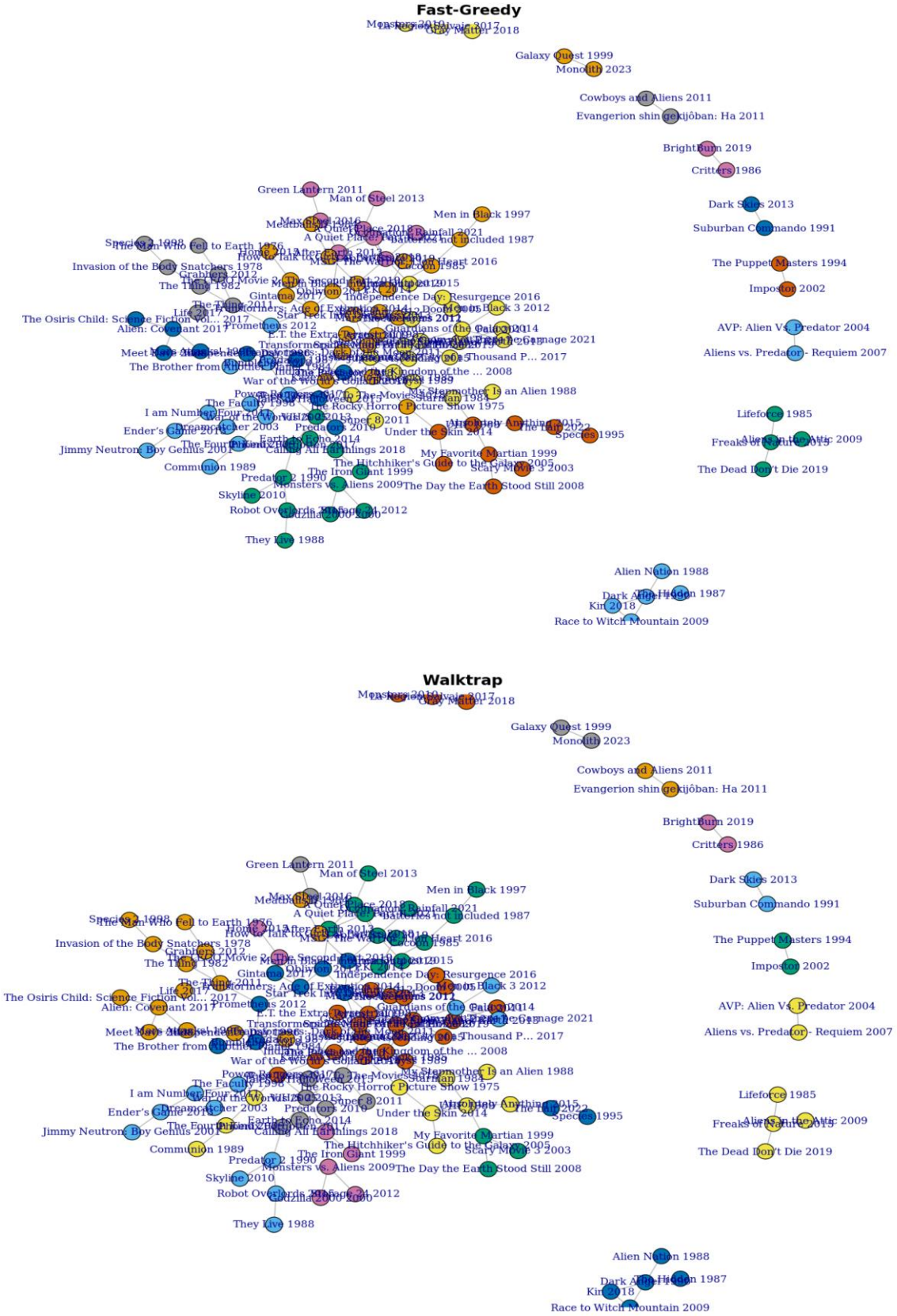
Zheng, Y. (2023). Improving box office projections through sentiment analysis: Insights from regularization-based forecast combinations. Economic Modelling, 125, 106349.	sentiment analysis improve box office projections?	based forecast combinations Data from the North American film market - October 1, 2010, to June 30, 2013.	Variables: Sentiment variables, movie characteristics; Dependent Variable: Box office projections	emphasizes the importance of sentiment variables, which consistently rank as top predictors across various forecast combination methods. - The Ridge and elastic net weighting schemes outperform other methods.
Eliashberg, J., Swami, S., Weinberg, C. B., & Wierenga, B. (2009). Evolutionary approach to the development of decision support systems in the movie industry. Decision Support Systems, 47(1), 1-12.	To explore the development and impact of the SilverScreener Decision Support System (DSS) in optimizing movie scheduling and revenue in the movie exhibition industry.	Regression analysis, quasi-experimental design	Independent variables: movie types, movie theater, week number. Dependent variables: revenue, net margin.	Significant coefficients identified through regression analysis. Substantial improvement in net margin observed over a six-month period, translating to an annual increase of over $900,000. Management deviations from DSS recommendations did not significantly affect net margin improvement, indicating the robustness of the system. DSS demonstrated efficacy in optimizing revenue and scheduling in the movie exhibition industry, with potential applications in the broader entertainment sector.
Hur, M., Kang, P., & Cho, S.	How can box-office forecasting	Experimentation with ISM and sentiment analysis	Independent variables: sentiment	ISM outperformed traditional methods in identifying

(2016). Box-office forecasting based on sentiments of movie reviews and Independent subspace method, Information Sciences, 372, 608-624.	accuracy be improved using sentiment analysis of movie reviews and the Independent Subspace Method (ISM)?	on movie review data	analysis results, Dependent variable: box-office performance	significant variables, sentiment factors improved forecasting accuracy, audience-related factors play crucial role in box-office forecasting.
Gaikar, D. D., & Marakarkandy, B. (2015). Using Twitter data to predict the performance of Bollywood movies. Industrial Management & Data Systems, 115(9), 1604-1621.	Can Twitter data be utilized to forecast the success of Bollywood films?	Twitter data extraction, sentiment analysis using PLSA classification, FIS implementation for prediction	Independent variables: Twitter data (tweets), Sentiment scores, Actor/actress rating Dependent variables: Box office performance prediction	The study introduced the concept of a "hype factor" calculated from Twitter data, which played a crucial role in predicting the opening weekend box office collections of movies. Twitter data, when analyzed using sentiment analysis and FIS, can effectively predict Bollywood movie performance and opening weekend box office collections.
Fan, L., Zhang, X., & Rai, L. (2021). When should star power and eWOM be responsible for the box office performance? - An empirical study based on signaling theory.	Are star power and electronic Word-of-Mouth (eWOM) significant factors affecting box office performance in the movie industry over different stages of a movie's theatrical run.	Empirical approach, analyzing panel data of 80 movies.	Independent Variables: Star power, eWOM volume, eWOM valence, signaling environment. Dependent Variables: Box office performance.	The star power significantly influences box office performance during the opening week, while eWOM becomes increasingly important in later-run weeks. The signaling environment amplifies the positive impact of star power on box office performance, particularly when

Journal of Retailing and Consumer Services, 62, 102591.				audiences are confronted with numerous movie choices.
Zhang, C., Tian, Y. P., & Fan, Z. P. (2022). Forecasting the box offices of movies coming soon using social media analysis: A method based on improved Bass models. Expert Systems with Applications	How can social media analysis be utilized to forecast the box office revenues of upcoming movies?	Data - EnData and Sina Weibo, 2013 to 2016 and WOM data from microblogs before movie releases. Sentiment analysis, genetic algorithm for parameter estimation in fitting the Bass models.	Independent variables: sentiment values, consumers' RFM values, and moviegoers' degree of attention derived from microblog data. Dependent variables: box office revenues of movies.	The study demonstrates the effectiveness of the proposed models in forecasting movie box office revenues, outperforming traditional methods and existing studies. Parameter estimation using genetic algorithms yields more accurate results than traditional methods. Social media analysis, particularly sentiment analysis, proves to be valuable in predicting movie revenues.
Hofmann, J., Clement, M., Völckner, F., & Hennig-Thurau, T. (2017). Empirical generalizations on the impact of stars on the economic success of movies. International Journal of Research in Marketing, 34(2), 442-461.	What is the impact of stars on the economic success of movies?	Propensity score matching techniques to mitigate selection biases.	Independent Variables: Commercial star power, Artistic star power; Dependent Variable: Box office revenues.	Movies featuring commercially successful stars exhibit a significant box office revenue premium, while those with artistically successful stars show a smaller and statistically insignificant revenue premium.

Zhang, L., Luo, J., & Yang, S. (2009). Forecasting box office revenue of movies with BP neural network. Expert Systems with Applications, 36(3), 6580-6587. https://doi.org/10.1016/j.eswa.2008.07.064	Can box office revenue of movies be forecasted accurately using a BP neural network?	Two years of box office data (2005-2006) from Wanda Cinema Line Company in China for 241 movies.	Independent variable: Movie characteristics (e.g., nationality, star value), competition data, cinema information. Dependent variable: Box office revenue	The BP neural network outperformed traditional MLP methods in forecasting box office revenue, achieving significantly better classification accuracy. The study underscores the stability, superior variable selection method, and robustness of the model, suggesting its potential application beyond movies.
Noakk, N., Nevolin, I., & Tatarnikov, A. (2012). Methodology for forecasting revenue from film distribution. Voprosy Ekonomiki, 48(138), 17-24.	Methodology for forecasting revenue from film distribution.	Factor analysis for identifying emotional factors influencing revenue. Linear regression model	Independent variables: Emotional evaluations, decision-making regarding movie attendance, scores on the Eysenck test. Dependent variable: Movie revenue.	No significant linear relationship between revenue and emotional evaluation overall. Some emotions like love influenced viewers' decision to attend movies. Identified three main factors influencing revenue: sadness, dynamism, and fear. Developed a linear regression model showing significant associations between revenue and emotional factors.

Figure 1. *Community detection algorithms on film graphs*



Appendix 3

Figure 1. *Hierarchical clustering methods*

Figure 1.1. *Average linkage on level 3.5*

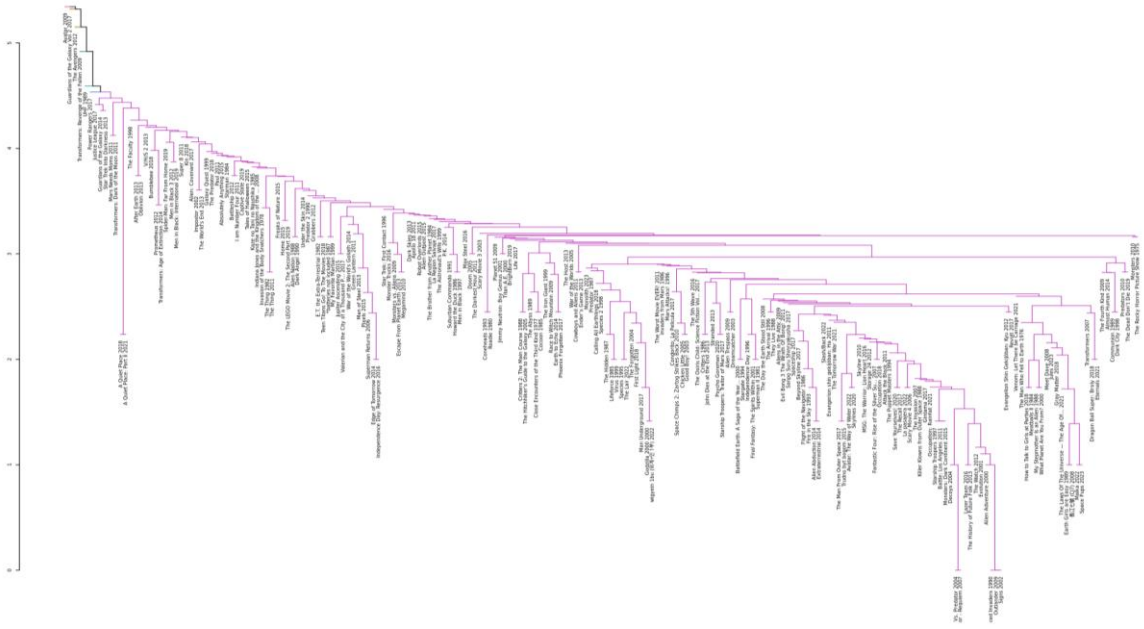


Figure 1.2. *Single linkage on level 2*

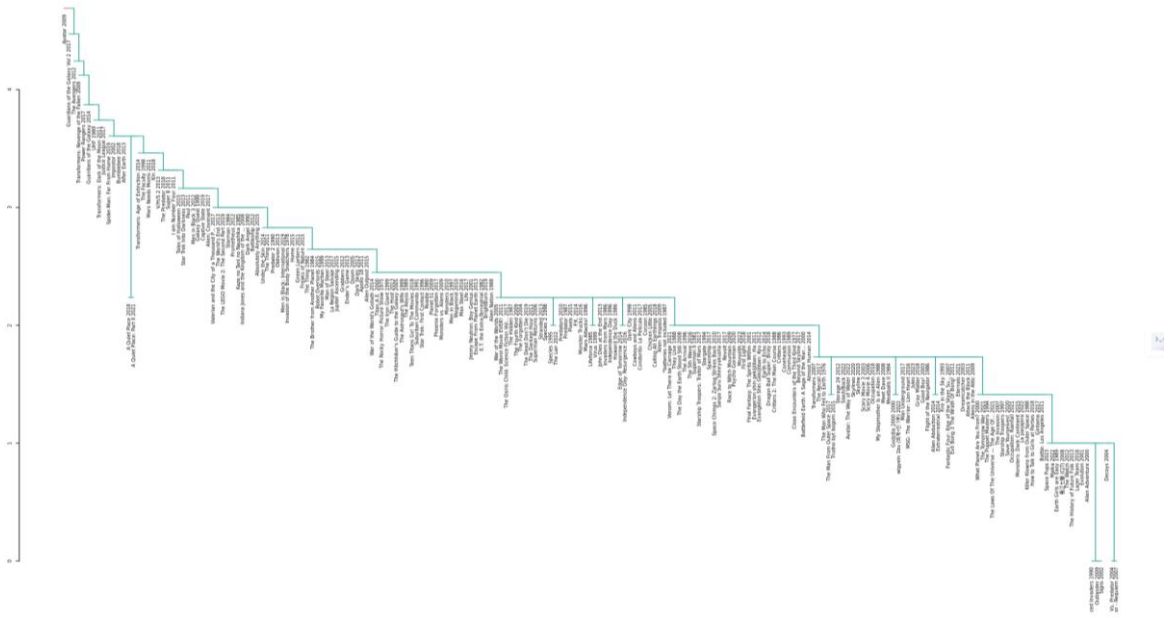


Figure 1.3. *Centroid linkage on level 1*

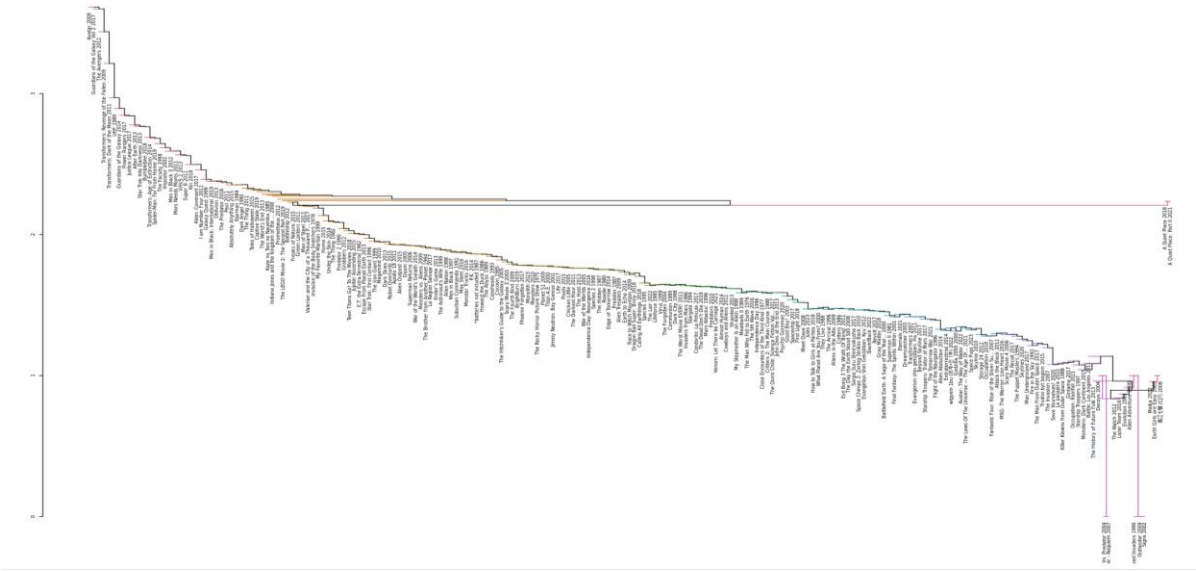
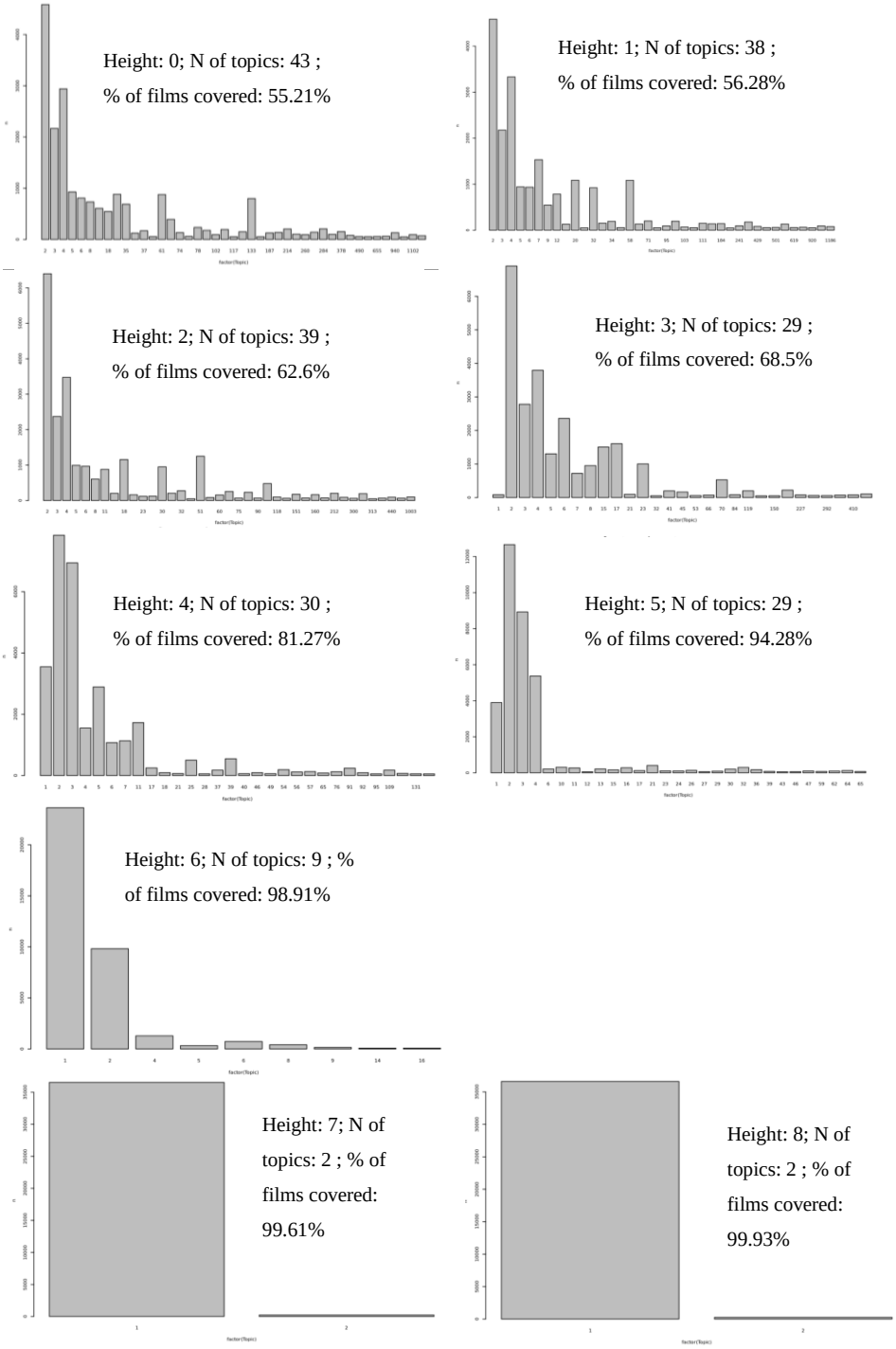


Figure 1. *Complete linkage level detection*



Appendix 5

Econometrics Estimations

	Dependent variable:	
	`Worldwide Box Office`	
	Real values	Log values
	(1)	(2)
Production	2.376***	0.813***
	(0.077)	(0.028)
MonthAugust	-20,577,259.000*	0.179
	(12,288,275.000)	(0.114)
MonthDecember	6,344,455.000	0.351***
	(11,885,087.000)	(0.111)
MonthFebruary	-15,500,892.000	0.129
	(13,086,573.000)	(0.122)
MonthJanuary	-21,710,635.000	0.151
	(14,343,824.000)	(0.134)
MonthJuly	5,650,767.000	0.260**
	(12,632,243.000)	(0.117)
MonthJune	9,276,196.000	0.299**
	(12,581,316.000)	(0.117)
MonthMarch	-11,925,571.000	0.161
	(12,607,576.000)	(0.117)
MonthMay	21,410,023.000	0.290**
	(13,051,843.000)	(0.121)
MonthNovember	-7,178,302.000	0.084
	(12,184,941.000)	(0.114)
MonthOctober	-17,264,259.000	0.002
	(12,071,184.000)	(0.112)

MonthSeptember	-17,334,117.000	0.085
	(12,606,087.000)	(0.118)
SourceBased on Factual Book/Article	-11,912,002.000	0.164
	(19,182,987.000)	(0.178)
SourceBased on Fiction Book/Short Story	-14,604,667.000	0.190
	(14,406,149.000)	(0.133)
SourceBased on Folk Tale/Legend/Fairytale	-11,131,649.000	0.546**
	(28,728,744.000)	(0.268)
SourceBased on Movie	-20,403,050.000	-0.196
	(32,280,771.000)	(0.300)
SourceBased on Musical or Opera	-163,391,720.000***	-0.007
	(46,480,219.000)	(0.433)
SourceBased on Play	-5,261,058.000	0.693***
	(27,467,820.000)	(0.255)
SourceBased on Real Life Events	-21,926,345.000	0.114
	(17,759,292.000)	(0.164)
SourceBased on smth else	-9,375,273.000	0.261
	(18,211,892.000)	(0.169)
SourceBased on TV	-47,480,530.000**	0.159
	(19,752,353.000)	(0.183)
SourceCompilation	-68,873,296.000	0.813
	(147,297,014.000)	(1.369)
SourceOriginal Screenplay	4,153,692.000	0.243*
	(13,832,215.000)	(0.128)
SourceRemake	-3,404,512.000	0.261
	(18,018,829.000)	(0.167)
GenreAdventure	26,662,723.000**	-0.088
	(12,613,491.000)	(0.117)

GenreBlack Comedy	5,485,440.000	0.126
	(19,378,740.000)	(0.180)
GenreComedy	-4,105,969.000	0.006
	(11,489,025.000)	(0.106)
GenreConcert/Performance	46,390,374.000	0.622
	(146,801,237.000)	(1.373)
GenreDocumentary	100,821,060.000	0.031
	(89,406,194.000)	(0.838)
GenreDrama	-461,816.000	-0.169
	(11,436,734.000)	(0.107)
GenreHorror	21,058,242.000	0.549***
	(14,968,578.000)	(0.140)
GenreMultiple Genres	-69,608,385.000	-1.088
	(103,925,354.000)	(0.968)
GenreMusical	116,495,020.000***	0.116
	(29,067,444.000)	(0.271)
GenreRomantic Comedy	14,818,487.000	0.002
	(20,418,720.000)	(0.190)
GenreThriller/Suspense	-12,468,152.000	-0.058
	(12,435,911.000)	(0.115)
GenreWestern	-43,012,707.000*	-0.282
	(25,432,648.000)	(0.237)
MPAANC-17	-55,905,301.000	0.472
	(87,187,546.000)	(0.814)
MPAAPG	-33,055,033.000*	-0.314*
	(20,000,133.000)	(0.187)
MPAAPG-13	-39,473,173.000*	-0.584***
	(20,586,556.000)	(0.192)

MPAAR	-52,379,913.000**	-0.901***
	(20,906,500.000)	(0.196)
`Running Time`	1,277,122.000***	1.396***
	(160,644.000)	(0.178)
Lead_power	19,189.000**	0.012
	(7,963.000)	(0.013)
Support_Power	32,944.000***	0.005
	(4,877.000)	(0.018)
Special_power	370,607.000	0.030
	(277,799.000)	(0.028)
Producer_Power	-16,571.000***	0.114***
	(5,854.000)	(0.013)
N_countries	1,416,701.000***	0.228***
	(167,380.000)	(0.023)
Prod_country_Power	-1,763.000***	-0.095***
	(304.400)	(0.025)
Topic109	-55,923,042.000	-0.784
	(79,095,006.000)	(0.737)
Topic11	-61,320,090.000	-0.566
	(70,664,993.000)	(0.659)
Topic129	50,673,918.000	-0.265
	(77,505,516.000)	(0.723)
Topic131	61,410,363.000	-0.286
	(76,024,949.000)	(0.709)
Topic17	1,768,912.000	-0.754
	(75,472,891.000)	(0.704)
Topic18	-13,313,406.000	0.056
	(75,639,210.000)	(0.705)

Topic183	-61,310,112.000	-0.444
	(76,132,160.000)	(0.710)
Topic2	19,058,394.000	-0.920
	(70,003,326.000)	(0.653)
Topic25	-11,958,175.000	-0.094
	(70,931,115.000)	(0.662)
Topic28	24,395,009.000	0.239
	(79,571,150.000)	(0.743)
Topic3	5,942,571.000	-0.452
	(68,765,889.000)	(0.641)
Topic37	-9,865,897.000	-0.625
	(72,082,379.000)	(0.672)
Topic39	61,462,115.000	-0.101
	(217,076,353.000)	(2.027)
Topic4	48,159,785.000	-0.489
	(70,826,343.000)	(0.661)
Topic40	14,700,538.000	0.544
	(94,674,659.000)	(0.883)
Topic46	-4,707,263.000	0.044
	(76,046,031.000)	(0.709)
Topic49	2,673,232.000	-0.148
	(86,162,481.000)	(0.804)
Topic5	11,547,692.000	-0.356
	(69,822,451.000)	(0.651)
Topic54	22,159,544.000	-0.355
	(73,596,812.000)	(0.686)
Topic56	-9,112,265.000	-0.653
	(75,762,261.000)	(0.706)

Topic57	-33,758,381.000	-0.121
	(71,180,638.000)	(0.664)
Topic6	6,228,994.000	-0.647
	(69,366,191.000)	(0.647)
Topic65	12,538,981.000	0.187
	(75,940,542.000)	(0.708)
Topic7	12,082,509.000	-0.199
	(71,523,661.000)	(0.667)
Topic76	26,373,499.000	0.206
	(72,960,687.000)	(0.681)
Topic91	-9,829,829.000	-1.176
	(88,792,720.000)	(0.828)
Topic92	-5,574,499.000	0.114
	(73,153,183.000)	(0.682)
Topic95	-158,824,114.000*	-1.658**
	(89,719,488.000)	(0.836)
TopicOthers	13,325,536.000	-0.171
	(68,152,235.000)	(0.636)
sentiment	3,262,227.000**	-0.015
	(1,539,899.000)	(0.014)
sentiment2	81,461.000	0.001
	(130,032.000)	(0.001)
anger	-1,485,655.000	-0.103***
	(2,138,385.000)	(0.020)
anticipation	2,589,524.000	-0.022
	(1,847,656.000)	(0.017)
disgust	5,140,356.000**	0.052**
	(2,320,377.000)	(0.022)

fear	-4,438,664.000**	0.016
	(1,976,893.000)	(0.018)
joy	-5,731,429.000**	0.051**
	(2,322,540.000)	(0.022)
negative	-36,281.000	-0.021
	(1,779,164.000)	(0.017)
positive	-1,536,563.000	-0.031**
	(1,652,599.000)	(0.015)
sadness	3,797,539.000*	-0.047**
	(2,009,560.000)	(0.019)
surprise	5,603,382.000**	0.050**
	(2,208,942.000)	(0.021)
trust	5,623,992.000***	0.021
	(1,729,140.000)	(0.016)
is_franchise1	112,528,570.000***	0.947***
	(6,971,265.000)	(0.065)
n_franchise	203,635.000	-0.049***
	(1,812,532.000)	(0.017)
Constant	-103,380,834.000	-2.123**
	(74,425,527.000)	(1.070)
Observations	3,556	3,556
R2	0.617	0.605
Adjusted R2	0.607	0.595
Residual Std. Error (df = 3465)	145,046,839.000	1.352
F Statistic (df = 90; 3465)	62.030***	58.920***
Note:	*p<0.1; **p<0.05; ***p<0.01	

Model 1: Real Values

Dependent Variable: Worldwide Box Office

Observations: 3,556

R-squared: 0.617

Adjusted R-squared: 0.607

Residual Std. Error: 145,046,839

F Statistic: 62.030 (significant at p < 0.01)

Significant Variables:

Production: Coefficient = 2.376 (p < 0.01)

Various Months

SourceBased on Musical or Opera: Coefficient = -163,391,720.000 (p < 0.01)

GenreMultiple Genres: Coefficient = -69,608,385.000 (p < 0.05)

Running Time: Coefficient = 1,277,122.000 (p < 0.01)

Lead_power: Coefficient = 19,189.000 (p < 0.01)

Support_Power: Coefficient = 32,944.000 (p < 0.01)

Producer_Power: Coefficient = -16,571.000 (p < 0.01)

N_countries: Coefficient = 1,416,701.000 (p < 0.01)

Prod_country_Power: Coefficient = -1,763.000 (p < 0.01)

is_franchise1: Coefficient = 112,528,570.000 (p < 0.01)

Model 2: Log Values

Dependent Variable: Log(Worldwide Box Office)

Observations: 3,556

R-squared: 0.605

Adjusted R-squared: 0.595

Residual Std. Error: 1.352

F Statistic: 58.920 (significant at p < 0.01)

Significant Variables:

Production: Coefficient = 0.813 (p < 0.01)

Various Months

SourceBased on Musical or Opera: Coefficient = -0.007 (p > 0.1)

GenreMultiple Genres: Coefficient = -1.088 (p < 0.01)

Running Time: Coefficient = 1.396 (p < 0.01)

Lead_power: Coefficient = 0.012 (p < 0.01)

Support_Power: Coefficient = 0.005 (p < 0.01)

Producer_Power: Coefficient = 0.114 (p < 0.01)

N_countries: Coefficient = 0.228 (p < 0.01)

Prod_country_Power: Coefficient = -0.095 (p < 0.01)

is_franchise1: Coefficient = 0.947 (p < 0.01)

Figure 1. *ML predictions*

Figure 1.1. *Real values prediction*

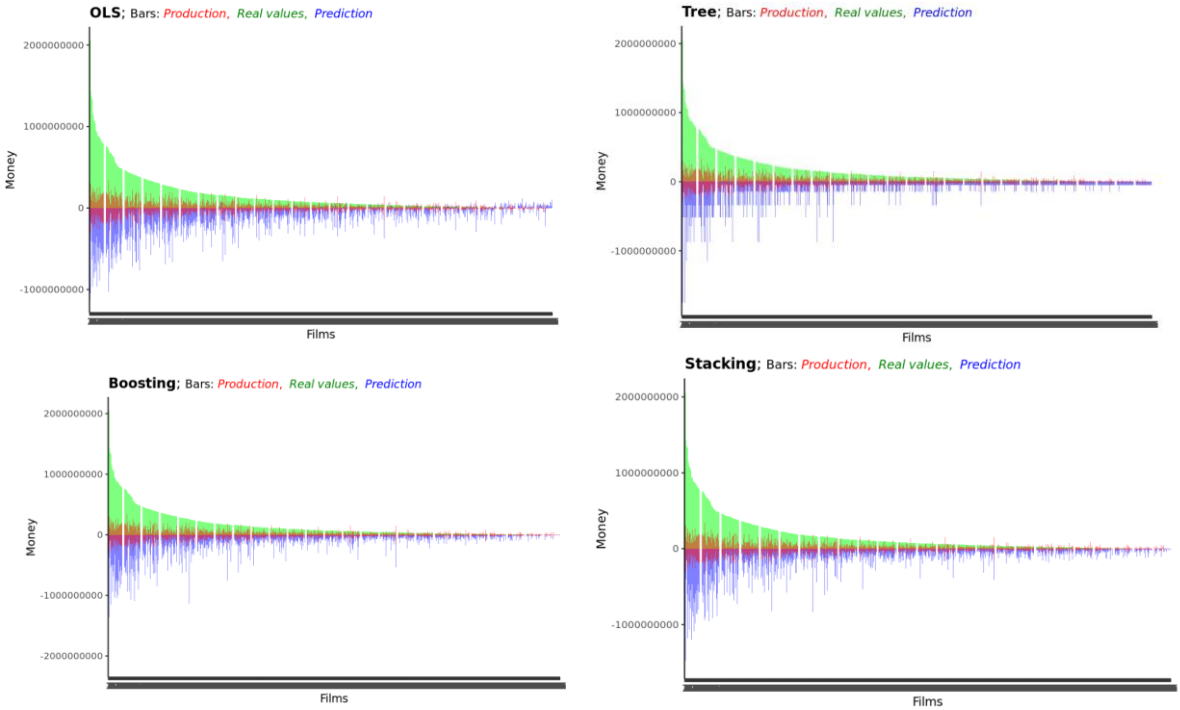
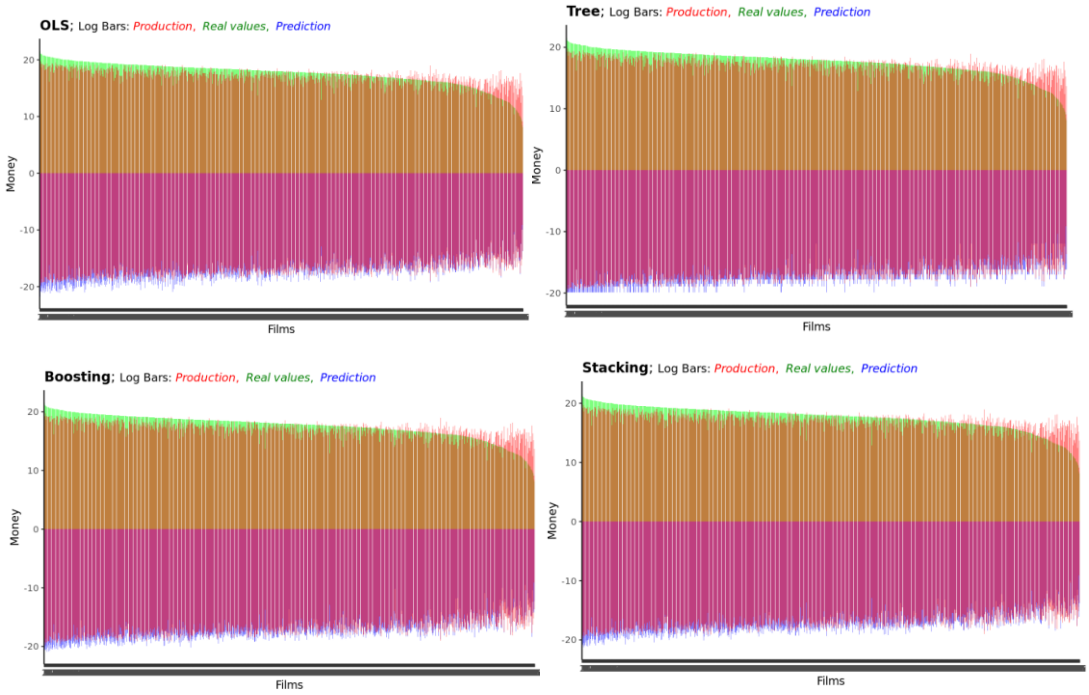



Figure 1.2. Logarithmic *values prediction*